

A mixture-of-experts deep learning architecture for daily streamflow prediction across the Sierra Nevada in California

Yu-Chieh Chao ^{a,b,*}, Wyatt Arnold^c, Minxue He^c, Zhaojun Bai^a, Zhi Ding^b, Prabhjot Sandhu^c, Peyman Namadi^c, Yiwei Cheng^c, Jamie Anderson^c, Shima Shamkhali Chenar^c, Elissa Lynn^c, K. A. Naditha Macaluso^c, Alejandro Perez^c, Andrew Schwarz^c

^a Department of Computer Science, University of California, Davis, California, USA

^b Department of Electrical and Computer Engineering, University of California, Davis, California, USA

^c California Department of Water Resources, Sacramento, California, USA

ARTICLE INFO

Keywords:

Deep learning
Mixture of experts
Long short-term memory
Streamflow prediction
Sierra Nevada
California

ABSTRACT

Reliable daily streamflow prediction is essential for water resources management in California's Sierra Nevada, where increasing hydroclimatic variability has intensified flood and drought risks. Although regional machine learning models can improve generalization across watersheds, conventional shared architectures may not adequately capture watershed-specific characteristics and distinct flow regimes. To address this limitation, we propose MoE- τ , a mixture-of-experts framework that combines multiple LSTM experts through an attention-based temporal gating network with a learnable temperature mechanism for adaptive expert weighting. We evaluated MoE- τ across 14 heterogeneous Sierra Nevada watersheds against learning-based models and the SAC-SMA process-based model. Under the in-domain temporal prediction benchmark, MoE- τ attained the highest mean NSE (0.873) among the learning-based models, a modest improvement over the strongest recurrent baseline (Bi-LSTM, 0.857) and a larger margin over SAC-SMA (0.762). It also remained robust under an alternative train-test split. Ablation and parameter-scale analyses showed that the performance gains arise from the proposed gating and expert-based design rather than increased model size. Model interpretation analyses revealed expert specialization across flow regimes and a strong relationship between hydrometeorological distribution differences and expert-usage differences (Spearman's $r = 0.687$). In held-out watershed experiments, MoE- τ outperformed LSTM on nearly all metrics across watersheds with distinct hydrologic conditions. These results demonstrate that adaptive expert selection improves both in-domain prediction and generalization to unseen watersheds, providing a robust and interpretable method for regional streamflow modeling.

* Corresponding author at: Department of Computer Science, University of California, Davis, California, USA.

E-mail address: ycjchao@ucdavis.edu (Y.-C. Chao).

<https://doi.org/10.1016/j.hydro.2026.100039>

Received 19 May 2026; Received in revised form 23 July 2026; Accepted 26 July 2026

1. Introduction

1.1. Background

Streamflow is a central component of the hydrologic cycle. It underpins water-resource systems worldwide, governing how water is stored, conveyed, and allocated for flood control, reservoir operations, groundwater recharge, ecosystem management, and urban and agricultural supply (Oki & Kanae, 2006; Viessman et al., 1989). Reliable streamflow prediction is therefore essential for both real-time decisions and long-term planning under climatic and hydrologic uncertainty (Milly et al., 2005; Vörösmarty et al., 2010). This is particularly critical in California of the United States, whose highly variable hydroclimate (Dettinger et al., 2011) is managed through a vast network of dams, reservoirs, aqueducts, and pumping plants that redistribute water from the wetter, mountainous north to the drier, agricultural, and urban south (Becker et al., 1976; Sabet & Coe, 1986). Here, prediction accuracy directly affects the efficiency, equity, and resilience of water supply and flood protection (Delaney et al., 2020; He et al., 2016).

This study focuses on the headwater watersheds of California's Sierra Nevada, which drain into the state's two largest river systems: the Sacramento and San Joaquin River basins. Together, these basins generate most of California's surface-water runoff and are the primary source for the state's major water-supply systems. They also provide a demanding, representative testbed for streamflow modeling because their hydroclimatology varies markedly with elevation and latitude. The northern Sacramento River Basin comprises generally larger, lower- to mid-elevation watersheds that are comparatively wet, more rainfall-dominated, and produce the largest runoff volumes, whereas the southern San Joaquin River Basin consists of higher-elevation watersheds that are more snowmelt-dominated and relatively arid (Dettinger et al., 2011; Rhoades et al., 2018). This hydroclimatic gradient, detailed in Section 2, creates substantial variation in rainfall- and snowmelt-driven streamflow dynamics across the study region. It therefore provides a rigorous setting for evaluating the predictive robustness and generalizability of data-driven models across heterogeneous watersheds.

1.2. Literature review

Streamflow prediction has traditionally followed two paradigms. Mechanistic (process-based) models such as the Sacramento Soil Moisture Accounting (SAC-SMA) model (Burnash, 1973) represent infiltration, evaporation, and storage through interconnected conceptual reservoirs. They are interpretable and remain standard in operational forecasting but demand substantial calibration, extensive data, and expert judgment (Duan et al., 1992; Hogue et al., 2000). Statistical models such as ARMAX (Box et al., 2015) instead fit empirical input–output relationships; they are simple and computationally efficient (Young & Wallis, 1985) but, by assuming linearity and stationarity, struggle to capture the nonlinear, time-varying nature of rainfall–runoff dynamics.

Machine learning (ML), and artificial neural networks (ANNs) in particular, emerged to overcome these limitations by learning nonlinear input–output mappings directly from data without prior structural assumptions. The seminal application by Hsu et al. (1995) showed that a three-layer feedforward ANN outperformed both SAC-SMA and ARMAX on the Leaf River Basin in Mississippi of the United States across multiple metrics and remained accurate under both wet and dry conditions. However, feedforward ANNs treat inputs independently and cannot retain information from previous time steps—a critical shortcoming for the highly autocorrelated hydrologic processes governed by antecedent conditions.

Recurrent architectures, especially Long Short-Term Memory (LSTM) networks (Hochreiter & Schmidhuber, 1997), addressed this gap through memory cells and gating that preserve temporal context, and now dominate data-driven streamflow modeling (Frame et al., 2021a,b; Hunt et al., 2022; Kratzert et al., 2018, 2019; Molina et al., 2025; Wilbrand et al., 2023). Kratzert et al. (2018) showed that LSTMs trained on 241 CAMELS catchments outperformed the coupled Snow-17/SAC-SMA model, especially in snow-influenced basins, while regional training enabled predictions in ungauged basins and fine-tuning further improved skill. Their main weaknesses were low-flow performance and the absence of explicit physical constraints. Subsequent work confirmed strong performance across gauged and ungauged basins and under nonstationarity (Arsenault et al., 2023; Kratzert et al., 2019). LSTMs have been applied less often to California watersheds, and the few existing studies expose recurring limitations. Wi and Steinschneider (2022) found that a regional LSTM reproduced expected seasonal runoff shifts under warming but could yield physically implausible increases in total runoff, which process-informed hybrid inputs only partly corrected. Adera et al. (2024) showed that LSTM post-processing of process-model outputs consistently improved accuracy while remaining insensitive to the calibration and complexity of the underlying model, yet still degraded during dry years and low flows. Xiao et al. (2025) used an LSTM post-processor over 23 Sierra Nevada basins to improve monthly and seasonal forecasts, with the largest gains in snow-dominated southern watersheds. Together, these studies share three limitations: reduced skill under hydrologic extremes and low flows, uncertain physical consistency, and unresolved trade-offs between regional and single-basin training.

Beyond standard LSTMs, recent work has pursued architectures better suited to the highly correlated, multi-scale, and noisy nature of streamflow series. Attention- and Transformer-based models have been benchmarked against recurrent networks (Liu et al., 2025), and interpretable Transformer–Kolmogorov–Arnold Network hybrids have fused multi-source inputs to improve simulation (Ding et al., 2026). Temporal convolutional networks with multi-scale feature fusion, denoising, and uncertainty quantification have improved predictions for strongly autocorrelated records (He et al., 2026), and decomposition–ensemble frameworks have enhanced robustness under nonstationary runoff (Wu et al., 2023). Others have improved skill through multi-source data integration (Yang et al., 2025), examined input uncertainty (Kim et al., 2026), and extended deep learning to snow- and glacier-influenced watersheds (Aguayo et al., 2025). More recently, Mixture-of-Experts (MoE) architectures that combine specialized recurrent or Informer experts through learned gating have improved daily runoff prediction over single-model approaches (Rong et al., 2025; Yang & Chen, 2026).

Across these paradigms, a consistent picture emerges. Process-based models offer physical interpretability but require substantial calibration. Statistical models are computationally efficient but often cannot represent the nonlinear, nonstationary characteristics of rainfall–runoff processes. Standard ANNs capture nonlinear relationships but lack temporal memory, whereas LSTMs model temporal dependencies effectively yet still struggle under hydrologic extremes, low flows, and extrapolation beyond the training domain. Recent attention-, convolution-, and MoE-based architectures have improved accuracy and robustness, but important questions remain. In particular, little is known about (1) the influence of training strategy, static watershed attributes, and input-window length; (2) model performance across diverse flow conditions; (3) the mechanisms underlying MoE performance gains; and (4) the generalizability of multi-watershed models to unseen watersheds. Addressing these gaps is particularly important for operational water-resources planning, where improvements in both predictive skill and model transparency can increase confidence in data-driven decision support.

1.3. Scope of the current work

This study addresses the four research gaps identified above by developing and evaluating an advanced LSTM-based framework for daily streamflow simulation across California's major water-supply watersheds. Specifically, it introduces MoE- τ , a novel sequence-level Mixture-of-Experts architecture in which an Attentive Temporal Gating Network with temperature scaling adaptively assigns expert weights to entire input sequences. Unlike existing MoE streamflow models that perform time step-level gating or rely primarily on Informer experts (Rong et al., 2025; Yang & Chen, 2026), MoE- τ is designed to better accommodate California's contrasting rain- and snow-dominated hydrologic regimes. The proposed framework is benchmarked against standard deep-learning models and the widely used process-based SAC-SMA model to address the following research questions:

- (1) Among the evaluated alternatives, which training strategy, static watershed attributes, and meteorological input-window length provide a suitable configuration for regional streamflow modeling?
- (2) How does MoE- τ compare with recurrent neural-network and process-based models for in-domain temporal prediction across general-flow, high-flow, low-flow, and peak-event metrics?
- (3) Are its performance gains attributable to the expert-based design rather than parameter count, and what are the contributions of attentive temporal gating and learnable temperature scaling?
- (4) How does multi-watershed training affect transfer beyond the training distribution, and does MoE- τ improve prediction relative to a shared LSTM for hold-out watersheds with contrasting hydrologic conditions?

The remainder of the paper is organized as follows. Section 2 describes the experimental setup, including the study area, dataset, and evaluation metrics. Section 3 presents the recurrent neural network models employed in the study. Section 4 reports the results, Section 5 discusses them, and Section 6 concludes the paper.

2. Experimental setup

2.1. Hydrological dataset

This study focuses on 14 watersheds located on the western slope of the Sierra Nevada in California (see Fig. 1 & Table 1). These watersheds are distributed across three major hydrologic regions within the Central Valley: the Sacramento Region (SHA, ORO, YRS, FOL), the San Joaquin Region (MKM, NHG, NML, TLG, MRC, MIL), and the Tulare Region (PNF, TRM, SCC, ISB). Collectively, these watersheds constitute the headwaters of California's river systems that drain into the Central Valley and Sacramento-San Joaquin Delta, serving as the primary source of water supply for urban, agricultural, and environmental uses statewide.

The hydroclimatic regime of the Sierra Nevada is strongly influenced by both elevation gradients and the Mediterranean-type climate that dominates much of California (Dettinger et al., 2011). Precipitation in these watersheds occurs predominantly between November and April, delivered primarily through winter frontal storms originating over the Pacific Ocean. The amount and form of precipitation vary significantly with elevation: lower elevations receive mostly rainfall, while higher elevations accumulate substantial snowpack. Snowpack is a defining characteristic of the Sierra Nevada watersheds, serving as a natural reservoir that stores water during the winter months and gradually releases it as runoff in the spring and early summer (Rhoades et al., 2018). This delayed runoff is fundamental to sustaining water supplies during California's dry season, when precipitation is minimal and agricultural and municipal demands are highest.

Temperature gradients across the watersheds exert a strong control on the timing and magnitude of snow accumulation and melt. Warmer winter and spring temperatures have already been shown to shift a portion of precipitation from snow to rain (Barnett et al., 2005; Lynn et al., 2020). This shift reduces snowpack storage and accelerates runoff timing, leading to earlier spring peak flows and reduced summer baseflows (Das et al., 2009; Godsey et al., 2014; Kapnick & Hall, 2010; Luce & Holden, 2009). These hydroclimatic changes have profound implications for reservoir operations, flood control, hydropower generation, and ecosystem management. For example, earlier runoff compresses the window in which reservoirs must simultaneously manage flood risk from high inflows and conserve water for dry-season supply, thereby increasing operational challenges. Hydrologic variability in the Sierra Nevada watersheds is further amplified by interannual and decadal-scale climate variability. Wet years, often associated with strong atmospheric river events, can produce extreme floods (Dettinger et al., 2011), while prolonged droughts, such as those observed between 2012 and 2016 (He et al., 2017), can drastically curtail runoff and groundwater recharge. Both extremes strain California's water management system: floods threaten downstream communities and infrastructure, whereas droughts stress agricultural production, urban

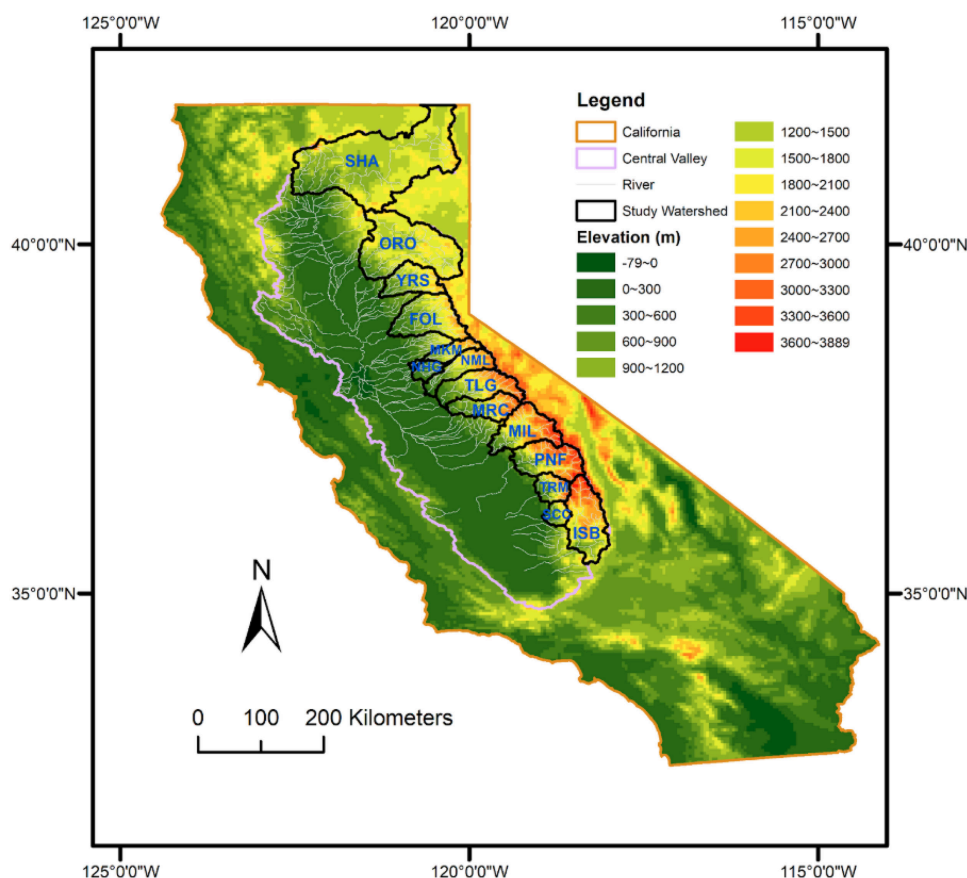


Fig. 1. Location map showing study watersheds in the Central Valley of California.

water supply reliability, and ecological habitats. The pronounced hydrologic variability makes predictive modeling and adaptive management critical components of statewide planning.

The spatial heterogeneity of hydroclimatic conditions across the Sacramento, San Joaquin, and Tulare regions is also of great significance. The Sacramento Region typically receives the highest annual precipitation, contributing to more reliable water yields relative to the southern regions. In contrast, the San Joaquin and especially the Tulare Region are more arid, with lower average precipitation, greater evapotranspiration demand, and higher interannual variability. As a result, water exported from the northern watersheds via the State Water Project (SWP) (Sabet & Coe, 1986) and Central Valley Project (CVP) (Becker et al., 1976) plays a disproportionately large role in supporting agriculture and urban uses in central and southern California. Understanding the hydroclimatic drivers across these regions is therefore central to forecasting water availability and managing statewide allocations.

In short, these 14 Sierra Nevada watersheds examined in this study embody the hydroclimatic diversity and variability that underpin California's water management challenges. Their snowpack-impacted regimes are sensitive to both natural climate variability and long-term climate change, making them essential case studies for advancing predictive hydrologic modeling and adaptive management strategies. A detailed understanding of their hydroclimatic characteristics not only improves scientific knowledge of watershed behavior but also informs policy and operational decisions aimed at securing a reliable and resilient water supply for the state.

For this study, observed daily streamflow records for water years 1987–2018 were obtained from the California Data Exchange Center (CDEC; <https://cdec.water.ca.gov/>). Daily precipitation and temperature data, including maximum, minimum, and average temperature, were downloaded at 4-kilometer resolution from the PRISM Climate Group (<http://www.prism.oregonstate.edu/>) and aggregated to the watershed scale for the same period. These hydroclimatic variables are complemented by static watershed attributes derived from three gridded datasets: soil classifications from the 1-km-resolution State Soil Geographic (STATSGO) dataset (Miller & White, 1998), land cover types from the 1-km-resolution Advanced Very High Resolution Radiometer (AVHRR) global land cover dataset (Hansen et al., 2010), and elevation from the Shuttle Radar Topographic Mission (SRTM) 90-m digital elevation model (Jarvis et al., 2008). These static attributes were aggregated to the watershed scale to characterize the dominant physiographic properties of each basin. Together, the dynamic hydroclimatic forcings and static watershed attributes form the core dataset for this study, providing a comprehensive representation of both time-varying climatic drivers and physiographic controls that are used to train and evaluate the ML models developed in this work.

Table 1
Physiographic and hydrologic characteristics of the 14 study watersheds.

ID	Name	Lon.	Lat.	Area	Elev.	Runoff	CV	1 April SWE [†]			
								2015 (dry)		2017 (wet)	
								Depth	Vol.	Depth	Vol.
		(°)	(°)	(km ²)	(m)	(mm yr ⁻¹)		(mm)	(BCM)	(mm)	(BCM)
SHA	Sacramento River above Shasta Lake	-122.42	40.718	17,262	1,404	396	0.39	7	0.120	115	1.983
ORO	Feather River above Lake Oroville	-121.493	39.54	9,342	1,546	553	0.52	3	0.026	198	1.853
YRS	Yuba River near Smartville	-121.274	39.235	2,870	1,445	941	0.54	4	0.011	399	1.144
FOL	American River above Folsom Lake	-121.183	38.683	4,882	1,347	722	0.56	7	0.035	339	1.653
MKM	Mokelumne River at Mokelumne Hill	-120.719	38.313	1,409	1,713	678	0.59	13	0.018	508	0.716
NHG	Calaveras River above New Hogan Lake	-120.812	38.152	940	639	234	0.65	0	0.000	0	0.000
NML	Stanislaus River at New Melones Reservoir	-120.525	37.948	2,331	1,735	564	0.64	19	0.044	506	1.179
TLG	Tuolumne River above La Grange Dam	-120.441	37.666	3,983	1,795	634	0.51	28	0.110	605	2.411
MRC	Merced River near Merced Falls	-120.331	37.522	2,748	1,647	420	0.68	13	0.036	440	1.211
MIL	San Joaquin River above Millerton Lake	-119.705	37.001	4,338	2,156	487	0.63	29	0.126	676	2.932
PNF	Kings River above Pine Flat Dam	-119.325	36.833	4,002	2,332	487	0.62	31	0.125	726	2.906
TRM	Kaweah River above Terminus Dam	-119.002	36.415	1,453	1,222	336	0.68	9	0.014	343	0.499
SCC	Tule River above Success Dam	-118.922	36.061	1,018	1,725	159	0.77	0	0.000	49	0.050
ISB	Kern River above Isabella Dam	-118.473	35.646	5,372	2,201	144	0.71	7	0.036	292	1.570

[†] Basin-average 1 April snow water equivalent (depth, and stored volume in billion cubic metres, BCM) derived from the Western United States UCLA Daily Snow Reanalysis (Fang et al., 2022). For each year, the 1 April SWE was averaged over each catchment. 2015 and 2017 are the lowest- and highest-snowpack years of the study period.

The physiographic and hydrologic characteristics of the 14 watersheds are summarized in Table 1. Catchment areas span 940–17,262 km² and mean elevations 639–2,332 m, reflecting the strong north-to-south gradient of the Sierra Nevada. Mean annual runoff, computed from the CDEC full-natural-flow records over 1987–2018, decreases from the wetter northern Sacramento basins (e.g. 941 mm yr⁻¹ at YRS) to the more arid southern Tulare basins (144 mm yr⁻¹ at ISB), and the interannual coefficient of variation of annual runoff rises correspondingly from 0.39 (SHA) to 0.77 (SCC), indicating greater year-to-year variability in the drier, more snow-dependent southern watersheds. The hydrologic importance of snow is underscored by the contrast between the driest and wettest years of the study period: averaged over the 14 basins, 1 April snow water equivalent stored only about 0.7 billion cubic metres (BCM) in 2015 but roughly 20 BCM in 2017, with the largest dry–wet swings in the high southern basins (e.g. the Kings River at PNF, 31 vs. 726 mm, and the San Joaquin River at MIL, 29 vs. 676 mm).

2.2. Process-based model benchmark

The PBM used in this study is the Sacramento Soil Moisture Accounting model (SAC-SMA; Burnash, 1973), a lumped conceptual hydrological model originally developed for the National Weather Service to produce river and flood forecasts across the United States (Burnash, 1995). SAC-SMA couples a 16-parameter soil moisture accounting module with auxiliary modules for potential evapotranspiration (PET) (Hamon, 1961), snow accumulation and melt (Snow-17; Anderson, 1973), and channel routing (Lohmann et al., 1998), yielding a total of 31 parameters. For each HRU i , Snow-17 converts precipitation into a rain-plus-melt input P_{eff} , and the SMA module partitions P_{eff} between actual evapotranspiration E_a , fast runoff Q_{dir} (impervious, surface, and interflow) and slow baseflow Q_{base} through the storage balance

$$\frac{dS_i}{dt} = P_{\text{eff}} - E_a - Q_{\text{dir},i} - Q_{\text{base},i}, \quad (1)$$

where S_i is the HRU storage state. The routing module then convolves the HRU runoff with a gamma unit hydrograph and a linearized Saint-Venant channel response and sums over HRUs to give simulated discharge $\hat{Q}(t) = \sum_i UH_i * (Q_{\text{dir},i} + Q_{\text{base},i})$. The full governing equations of the four modules are given in Appendix B.

We adopt, as the PBM benchmark, the spatially distributed SAC-SMA model that was developed and calibrated for these Sierra Nevada watersheds by Wi and Steinschneider (2022); the calibrated model is used here without modification. In that framework the model is built on over 6,000 HRUs, delineated by intersecting approximately 6 km climate grid cells with soil classes from the STATSGO dataset (Miller & White, 1998); elevation and vegetation are upscaled to the HRU level from the SRTM 90 m digital elevation model (Jarvis et al., 2008) and 1 km AVHRR land-cover data (Hansen et al., 2010), and HRU temperatures are adjusted using monthly lapse rates derived from MODIS land-surface temperature (Wan, 2014). SAC-SMA has 31 parameters in total (Table B.12). To avoid over-parameterization, Wi and Steinschneider (2022) tied the soil-moisture-accounting parameters to HRUs of the same STATSGO soil class (12 classes) and the Hamon PET coefficient to HRUs of the same AVHRR vegetation class (11 classes), lumped the Snow-17 and routing parameters across all watersheds, and held side, SCF, and PXTMP fixed at the values in Table B.12, giving 203 calibrated parameters. They calibrated the model with the genetic algorithm of Wang (1991) under the pooled, multi-basin scheme of Wi et al. (2015), which maximizes the average Kling–Gupta Efficiency (KGE) across all gauges simultaneously, over a calibration period of 1989–2003 with validation over 2004–2018. Further details are given in Appendix B.

2.3. Evaluation metrics

Model performance is assessed using nine metrics computed with the evaluation functions provided by the NeuralHydrology library (Kratzert et al., 2022). These metrics summarize both overall predictive skill and performance in hydrologically important flow regimes. The overall-fit metrics include the Nash-Sutcliffe Efficiency (NSE) (Nash & Sutcliffe, 1970), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Kling-Gupta Efficiency (KGE) (Gupta et al., 2009), and Pearson correlation coefficient (Pearson- r). NSE is defined as:

$$\text{NSE} = 1 - \frac{\sum_{t=1}^T (\hat{y}^t - y^t)^2}{\sum_{t=1}^T (y^t - \bar{y})^2}, \quad (2)$$

where $\hat{y}$ and y denote simulated and observed discharge, respectively, and $\bar{y}$ is the mean observed discharge. MSE quantifies the average squared prediction error:

$$\text{MSE} = \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t)^2. \quad (3)$$

RMSE is the square root of MSE and reports the error measure in the original discharge units. KGE further decomposes performance into correlation, variability bias, and mean bias components:

$$\text{KGE} = 1 - \sqrt{(r - 1)^2 + (\alpha - 1)^2 + (\beta_{\text{KGE}} - 1)^2}, \quad (4)$$

where r is the Pearson correlation coefficient, α is the ratio of simulated to observed standard deviations, and β_{KGE} is the ratio of simulated to observed means. KGE was also used as the objective function for calibrating the PBM.

Extreme-flow metrics are used to assess model reliability under hydrologically important high- and low-flow conditions. The percent bias in high-flow volume (FHV) (Yilmaz et al., 2008) measures bias in the upper tail of the flow duration curve. In this study, the high-flow segment is defined as the top 2% of flows:

$$\% \text{BiasFHV} = \frac{\sum_{h=1}^H (\hat{y}_h - y_h)}{\sum_{h=1}^H y_h} \times 100, \quad (5)$$

where h indexes the selected high-flow values, and H is the number of values included in the upper 2% of the flow duration curve. Similarly, the percent bias in low-flow volume (FLV) (Yilmaz et al., 2008) evaluates the bottom portion of the flow duration curve using a logarithmic formulation that emphasizes relative errors in the low-flow regime:

$$\% \text{BiasFLV} = -1 \times \frac{\sum_{l=1}^L [\log(\hat{y}_l) - \log(y_L)] - \sum_{l=1}^L [\log(y_l) - \log(y_L)]}{\sum_{l=1}^L [\log(y_l) - \log(y_L)]} \times 100 \quad (6)$$

where l indexes the selected low-flow values, and L is the number of values included in the lower 30% of the flow duration curve.

Peak flow characteristics are further evaluated using Peak-MAPE (Kratzert et al., 2022) and Peak-Timing (Kratzert et al., 2021) metrics. Peak-MAPE measures the mean absolute percentage error of peak flow magnitudes at observed peak-flow events:

$$\text{MAPE}_{\text{peak}} = \frac{1}{P} \sum_{p=1}^P \left| \frac{\hat{y}_p - y_p}{y_p} \right| \times 100, \quad (7)$$

where P is the number of observed peaks identified from the observed discharge series using the SciPy Python package's `find_peaks` algorithm, with the minimum horizontal distance set to 100 days (i.e., at least 100 days of separation between candidate peaks). Here, y_p denotes the observed discharge at the p -th observed peak, and $\hat{y}_p$ denotes the corresponding simulated discharge at the same time index. Peak-Timing quantifies the mean absolute time difference (in days) between observed and simulated peak occurrences.

For all metrics, optimal performance is indicated by values of 1 for NSE, KGE, and Pearson- r and by values approaching 0 for MSE, RMSE, FHV, FLV, Peak-MAPE, and Peak-Timing. While the overall-fit metrics (NSE, MSE, RMSE, KGE, Pearson- r) provide a comprehensive assessment of general model capacity across typical flow conditions, the extreme-flow metrics (FHV, FLV, Peak-MAPE, Peak-Timing) offer targeted diagnostics essential for evaluating model reliability under the high-flow events relevant to flood management and the low-flow conditions critical for water supply planning and drought assessment.

3. Method

To assess the capacity of recurrent neural network architectures to model static watershed attributes and dynamic data for stream-flow discharge prediction, we conducted a comprehensive set of experiments spanning Gated Recurrent Unit (GRU) (Cho et al., 2014), unidirectional LSTM (LSTM) (Graves, 2012), bidirectional LSTM (Bi-LSTM) (Schuster & Paliwal, 1997), LSTM with Attention mechanism (LSTM-Attn) (Bahdanau et al., 2014), Convolutional LSTM (CNN-LSTM) (Shi et al., 2015), Mixture-of-Experts (MoE) (Jacobs et al., 1991), and the proposed MoE- τ .

The dataset spans 14 watersheds, and the model is trained in a per-watershed single-input single-output setting. Each sample is a tuple $(x_{1:T}, s, y)$, where $x_{1:T} \in \mathbb{R}^{T \times D_d}$ denotes the meteorological forcings over the past T days, with each day represented by a vector $x_t \in \mathbb{R}^{D_d}$ of D_d dynamic forcings, $s \in \mathbb{R}^{D_s}$ is the vector of static attributes of the watershed, and $y \in \mathbb{R}$ is the target discharge

of day T . We condition on s at every timestep by broadcasting and concatenation, $\tilde{x}_t = [x_t; s] \in \mathbb{R}^{D_d+D_s}$, $\forall t \in \{1, \dots, T\}$, yielding $\tilde{x}_{1:T} \in \mathbb{R}^{T \times (D_d+D_s)}$. The model then predicts discharge via $\hat{y} = f_\theta(\tilde{x}_{1:T})$, where f_θ is the model with trainable parameters θ and $\hat{y} \in \mathbb{R}$ be the predicted discharge. Model f_θ can be GRU, LSTM, Bi-LSTM, LSTM-Attn, or MoE.

3.1. GRU

As a first baseline, we employ the GRU, a standard recurrent architecture for sequence modeling. Given the input sequence $\tilde{x}_{1:T}$, a single-layer GRU with hidden dimension D_h generates a sequence of hidden states $h_{1:T} = \text{GRU}_\phi(\tilde{x}_{1:T})$, where $h_t \in \mathbb{R}^{D_h}$ and ϕ denotes the GRU learnable parameters. For streamflow prediction, we use the final hidden state h_T as a summary of the input sequence and apply a linear probing $\hat{y} = w^\top h_T + b$, with w and b as learnable output layer parameters.

3.2. LSTM

Next, we consider the standard LSTM network, which augments the recurrent update with an explicit memory cell and three gates (input, forget, and output) to regulate how information is stored, discarded, and exposed over time. This design generally improves the ability to capture longer-range temporal dependencies as compared to GRU. Given the input sequence $\tilde{x}_{1:T}$, a LSTM with hidden dimension D_h generates hidden states $h_{1:T} = \text{LSTM}_\theta(\tilde{x}_{1:T})$, where $h_t \in \mathbb{R}^{D_h}$ and θ denotes the LSTM learnable parameters. The streamflow prediction is obtained via the same linear probing on the final hidden state $\hat{y} = w^\top h_T + b$.

3.3. Bi-LSTM

We also investigate Bi-LSTM which processes the input sequence in both forward and reverse temporal directions, allowing the final representation to exploit context from both earlier and later timesteps. Concretely, a forward LSTM generates hidden states $h_{1:T}^f = \text{LSTM}_{\theta_f}^f(\tilde{x}_{1:T})$ and a backward LSTM generates hidden states $h_{1:T}^b = \text{LSTM}_{\theta_b}^b(\tilde{x}_{1:T})$. For each timestep, we concatenate the two hidden states to get $h_t^{bi} = [h_t^f; h_t^b] \forall t \in [1, \dots, T]$, where $h_t^{bi} \in \mathbb{R}^{2D_h}$, and use the last hidden state for prediction $\hat{y} = w^\top h_T^{bi} + b$.

3.4. LSTM-Attn

We also consider incorporating LSTM with a temporal attention mechanism, named LSTM-Attn. Instead of using only the last hidden state of LSTM, we apply a self-attention layer over all hidden states $h_{1:T}$ to compute a weighted combination that emphasizes the most relevant timesteps for the current prediction.

3.5. CNN-LSTM

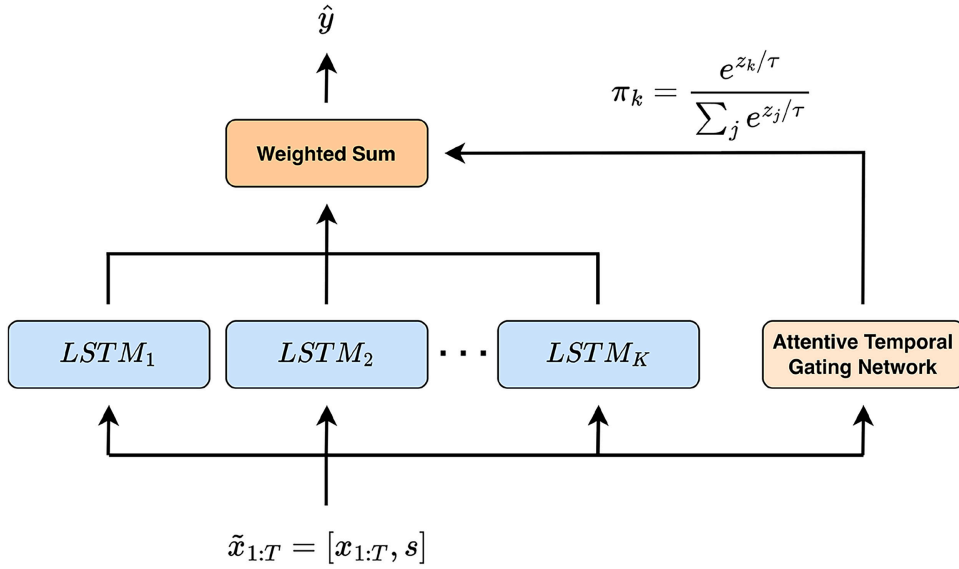
Next, we consider the CNN-LSTM architecture, which augments the LSTM with a 1D Convolutional Neural Network. The convolutional layer acts as a feature extractor over the input sequence, aggregating local temporal patterns and reducing noise before passing features to the recurrent module. Given the embedded input sequence $\tilde{x}_{1:T}$, the 1D convolutional layer transforms it into a pattern sequence $\tilde{z}_{1:T'}$, where each $\tilde{z}_t$ encodes local context over a short temporal window. This sequence is then processed by an LSTM with hidden dimension D_h , producing hidden states $h_{1:T'} = \text{LSTM}_\theta(\tilde{z}_{1:T'})$, where $h_t \in \mathbb{R}^{D_h}$ and θ denotes the learnable LSTM parameters. Similar to the pure LSTM model, the streamflow prediction is obtained by applying a linear head to the LSTM outputs, yielding the prediction $\hat{y}$.

3.6. MoE

Different watersheds exhibit distinct discharge distributions, hydrological responses, and temporal dynamics. Based on our experimental observations, it is challenging for a single recurrent model, such as an LSTM, to learn one shared representation space that generalizes well across all watershed-specific distributions. In particular, when heterogeneous watersheds are modeled jointly, a single LSTM must encode multiple potentially different input-output mappings within the same latent space, which can limit its ability to capture distribution-specific temporal patterns. Moreover, increasing model depth by stacking additional LSTM layers does not effectively address this limitation, suggesting that the main challenge is not only insufficient network capacity, but also the lack of an adaptive mechanism for handling heterogeneous watershed distributions.

Motivated by this, we adopt a Mixture-of-Experts (MoE) design inspired by the classical MoE framework (Jacobs et al., 1991). In a standard MoE architecture, multiple expert models are trained in parallel, and a gating network determines the contribution of each expert for a given input. In our setting, each expert is implemented as an independent LSTM, resulting in a Mixture-of-LSTMs architecture. This design allows different LSTM experts to learn specialized latent representations for different discharge distributions, watershed characteristics, or temporal predictive behaviors, rather than forcing all watershed dynamics to be represented by a single recurrent model.

To make the expert selection input-dependent and temporally informed, we propose an Attentive Temporal Gating Network for the sequence-level MoE. This gating network consists of an LSTM followed by a temporal attention mechanism. The LSTM first encodes the temporal structure of the input sequence, while the attention mechanism aggregates the hidden states into a sequence-level context representation that is used to estimate the expert weights. Since the final prediction is formed as a weighted combination

Fig. 2. Model architecture of the proposed MoE- τ .

of expert outputs, we further consider the possibility that a standard softmax gating function may produce overly smooth expert weights, making the prediction behave similarly to an average over multiple experts. This would be undesirable for our intended design, where different LSTM experts are expected to learn distribution-specific representations and contribute more strongly when the input sequence matches their specialized predictive behavior. To provide the model with more flexibility in controlling this behavior, we introduce a learnable temperature parameter in the gating softmax function. Instead of fixing the sharpness of the expert-weight distribution manually, the temperature is optimized jointly with the rest of the model to minimize the prediction loss. With this temperature-controlled Attentive Temporal Gating Network, we denote the proposed sequence-level Mixture-of-LSTMs model as MoE- τ .

Given the input sequence $\tilde{x}_{1:T}$, each expert LSTM independently encodes the sequence and produces hidden states,

$$h_{1:T}^{(k)} = \text{LSTM}_{\theta_k}^{(k)}(\tilde{x}_{1:T}), \quad k \in \{1, \dots, K\}, \quad (8)$$

where θ_k denotes the learnable parameters of expert k , K is the number of experts, and $h_t^{(k)} \in \mathbb{R}^{D_h}$ denotes the hidden state of expert k at timestep t . Since each expert has its own parameters, the experts are allowed to learn different temporal representations for different watershed distributions or predictive behaviors.

The Attentive Temporal Gating Network also receives the input sequence $\tilde{x}_{1:T}$ and determines the sequence-level contribution of each expert. The gating module first uses an LSTM to encode the temporal structure of the input sequence,

$$g_{1:T} = \text{LSTM}_{\theta_g}^{\text{gate}}(\tilde{x}_{1:T}), \quad (9)$$

where θ_g denotes the learnable parameters of the gating LSTM and $g_t \in \mathbb{R}^{D_g}$ denotes the gating hidden state at timestep t . To summarize these hidden states into a sequence-level context vector, we apply temporal attention,

$$u_t = v^\top \tanh(W g_t), \quad \alpha_t = \frac{e^{u_t}}{\sum_{j=1}^T e^{u_j}}, \quad (10)$$

where $W \in \mathbb{R}^{D_a \times D_g}$ and $v \in \mathbb{R}^{D_a}$ are learnable parameters, u_t is the attention score, and α_t is the normalized attention weight at timestep t . The gating context vector is then computed as the weighted sum of the gating hidden states,

$$c = \sum_{t=1}^T \alpha_t g_t, \quad (11)$$

where $c \in \mathbb{R}^{D_g}$ serves as a temporally attended sequence-level representation for expert selection. Based on the context vector c , we compute the expert logits,

$$z = W_p c, \quad (12)$$

where $z \in \mathbb{R}^K$, and $W_p \in \mathbb{R}^{K \times D_g}$ is a learnable projection matrix. Since the final prediction is formed from a weighted combination of expert representations, we introduce a learnable temperature parameter in the gating softmax function to give the model additional flexibility in controlling the sharpness of the expert-weight distribution. Instead of manually fixing this sharpness, the temperature is optimized jointly with the rest of the model to minimize the prediction loss. The expert weights are computed as

$$\pi_k = \frac{e^{z_k/\tau}}{\sum_{j=1}^K e^{z_j/\tau}}, \quad (13)$$

where $\pi \in \mathbb{R}^K$ denotes the sequence-level expert weights and τ is the learnable temperature parameter. A smaller temperature produces a sharper expert-weight distribution, allowing the model to rely more strongly on selected experts when beneficial for prediction.

The mixture representation is obtained as a convex combination of the experts' final hidden states,

$$m = \sum_{k=1}^K \pi_k h_T^{(k)}, \quad (14)$$

where $h_T^{(k)} \in \mathbb{R}^{D_h}$ and $m \in \mathbb{R}^{D_h}$. For streamflow prediction, we apply a linear prediction head,

$$\hat{y} = w^T m + b, \quad (15)$$

with $w \in \mathbb{R}^{D_h}$ and $b \in \mathbb{R}$ as learnable parameters. This sequence-level MoE- τ design, shown in Fig. 2, allows different LSTM experts to model different temporal patterns, while the Attentive Temporal Gating Network adaptively assigns expert weights according to each input sequence.

4. Experimental results

The experimental design follows a two-stage evaluation strategy. In the first stage, we assess the in-domain temporal predictive capability of different models by training them on the historical period of the study watersheds and evaluating their performance on future periods of the same watersheds. This stage identifies the model architecture that provides the strongest predictive performance under the standard temporal predictive setting. In the second stage, we examine whether the selected model can generalize beyond the watersheds used for training. Specifically, the model is evaluated under a held-out watershed setting, where one or more target watersheds are excluded from training and used only for testing. This two-stage design allows us to evaluate both predictive accuracy within observed watersheds and spatial generalization capacity to unseen watersheds.

This section presents the experimental evaluation in a step-by-step manner. Section 4.1 first summarizes the implementation details. Section 4.2 provides preliminary analyses for constructing a shared watershed modeling framework. Specifically, the specialized-versus-shared LSTM comparison motivates the use of one shared model across watersheds rather than training separate models for individual watersheds, since the goal is to develop a model that can be transferred to unseen watersheds. The feature selection analysis then determines the static attributes and input forcing-window length used in the subsequent experiments. Section 4.3 presents Stage I, which evaluates the in-domain temporal predictive performance of the proposed MoE- τ method against other learning-based models. Section 4.4 further analyzes MoE- τ from multiple perspectives. We first use parameter-scale comparisons to verify that the performance gain is not simply due to increased model capacity, and then conduct ablation studies to assess the contribution of each proposed component. Then, we provide model interpretation analyses to understand the learned behavior of the experts, including how their activation patterns vary across watershed-specific data distributions and different streamflow regimes. Finally, Section 4.5 presents Stage II, which evaluates the generalization capacity of MoE- τ under the held-out watershed setting, where the model is tested on watersheds excluded from training.

4.1. Implementation details

In-domain temporal predictive benchmark. For the in-domain temporal predictive experiments, each model is trained and tested using a temporal split within the study watersheds. Unless otherwise specified, the training period spans from 1987/10/01 to 2012/12/31, and the testing period spans from 2013/01/01 to 2018/12/31. The only exception is discussed in Table 7, where the train-test split is reversed.

Held-out watershed generalization. For the held-out watershed experiments, the split is performed by watershed rather than by time. Specifically, each selected target watershed is excluded from training, while the model is trained on the remaining watersheds over the full available period from 1987/10/01 to 2018/12/31. The trained model is then evaluated over the same period on the held-out watershed to assess its generalization capacity to unseen watersheds.

All neural network models are implemented in PyTorch (Paszke et al., 2019). The dynamic meteorological forcings used as model inputs include daily precipitation, minimum temperature, and maximum temperature, with precipitation measured in mm/day and temperature measured in °C. The prediction target is the discharge on the last day of the input window, which is converted from cubic feet per second (cfs) to runoff depth in mm/day. The training loss function is Mean-Squared Error (MSE). Detailed descriptions of the experimental settings and model configurations are provided in Appendix A. Specifically, the shared experimental hyperparameters are summarized in Table A.10, while the model-specific configurations are presented in Table A.11.

Table 2

In-domain Performance: Comparison of specialized and shared LSTM configurations on the SHA watershed. The specialized LSTM is trained solely on SHA, whereas the shared LSTM is jointly trained on SHA, NML, and TRM. The specialized LSTM achieves superior in-domain performance compared to the shared LSTM.

LSTM	NSE ($-\infty, 1$] $\uparrow$	MSE [0, ∞) $\downarrow$	RMSE [0, ∞) $\downarrow$	KGE ($-\infty, 1$] $\uparrow$	FHV $\rightarrow 0$	FLV $\rightarrow 0$	Pearson-r [-1, 1] $\uparrow$	Peak-MAPE [0, ∞) $\downarrow$	Peak-Timing $\rightarrow 0$
Specialized	0.861	0.222	0.471	0.794	-17.604	73.188	0.937	21.532	0.633
Shared	0.828	0.274	0.520	0.793	-19.951	73.039	0.917	28.256	0.400

Table 3

Generalization Capacity: Comparison of specialized and shared LSTM configurations on the unseen ISB watershed. The specialized LSTM is trained solely on SHA, whereas the shared LSTM is jointly trained on SHA, NML, and TRM. The shared LSTM exhibits superior generalization capacity on the unseen distribution.

LSTM	NSE ($-\infty, 1$] $\uparrow$	MSE [0, ∞) $\downarrow$	RMSE [0, ∞) $\downarrow$	KGE ($-\infty, 1$] $\uparrow$	FHV $\rightarrow 0$	FLV $\rightarrow 0$	Pearson-r [-1, 1] $\uparrow$	Peak-MAPE [0, ∞) $\downarrow$	Peak-Timing $\rightarrow 0$
Specialized	-0.692	0.587	0.765	-0.160	60.583	-117.580	0.512	72.779	0.650
Shared	0.698	0.105	0.323	0.409	0.105	-37.564	0.902	24.623	0.050

Table 4

Static attribute feature selection experiment. The – symbol indicates that no static attribute is used. One static attribute is added at a time to identify the most effective static attributes for subsequent experiments. The model used in this experiment is an MoE- τ with 8 experts.

Static Attributes s	NSE ($-\infty, 1$] $\uparrow$	MSE [0, ∞) $\downarrow$	RMSE [0, ∞) $\downarrow$	KGE ($-\infty, 1$] $\uparrow$	FHV $\rightarrow 0$	FLV $\rightarrow 0$	Pearson-r [-1, 1] $\uparrow$	Peak-MAPE [0, ∞) $\downarrow$	Peak-Timing $\rightarrow 0$
–	0.807	1.072	0.951	0.789	-8.337	-136.004	0.917	36.370	0.432
+ Elevation	0.850	0.888	0.853	0.805	-11.696	7.583	0.935	29.006	0.420
+ Land	0.869	0.824	0.818	0.830	-10.962	-22.962	0.938	28.720	0.416
+ Soil	0.859	0.875	0.845	0.830	-13.316	-10.679	0.931	30.751	0.392
+ Flow Length	0.851	0.905	0.860	0.812	-10.371	-61.528	0.932	30.301	0.411

Table 5

Experiments for selecting the input forcing-window length. The model used in this experiment is MoE- τ with 8 experts.

Window (days)	NSE ($-\infty, 1$] $\uparrow$	MSE [0, ∞) $\downarrow$	RMSE [0, ∞) $\downarrow$	KGE ($-\infty, 1$] $\uparrow$	FHV $\rightarrow 0$	FLV $\rightarrow 0$	Pearson-r [-1, 1] $\uparrow$	Peak-MAPE [0, ∞) $\downarrow$	Peak-Timing $\rightarrow 0$
30	0.472	2.539	1.504	0.579	-17.751	-135.054	0.719	47.950	0.538
180	0.847	0.968	0.887	0.796	-12.834	25.315	0.930	30.670	0.435
365	0.873	0.813	0.813	0.839	-9.475	-22.226	0.939	28.513	0.422
730	0.862	0.869	0.842	0.833	-12.274	-107.637	0.933	27.771	0.434

4.2. Preliminary analyses for shared watershed modeling

Before the main experimental evaluation, we conduct preliminary analyses to justify the shared watershed modeling framework and determine the features used in subsequent experiments.

Specialized vs. shared LSTM. We use a single-layer LSTM to compare specialized and shared watershed modeling strategies. Table 2 evaluates the in-domain setting, where the specialized LSTM is trained and tested on the SHA watershed, while the shared LSTM is trained jointly on SHA, NML, and TRM and tested on SHA. The results show that the specialized LSTM achieves slightly better in-domain performance, indicating the benefit of watershed-specific training when sufficient target-watershed data are available. Table 3 evaluates generalization to an unseen watershed, where the specialized LSTM is trained on SHA and tested on ISB, while the shared LSTM is trained on SHA, NML, and TRM and tested on ISB. In this setting, the shared LSTM achieves better performance across all evaluation metrics, showing that training across multiple watersheds improves generalization to an unseen watershed.

Overall, although the specialized LSTM provides slightly better in-domain performance, the shared LSTM still maintains reasonable performance in the in-domain setting while demonstrating better generalization to an unseen watershed. Since the model is expected to be applicable to unseen watersheds, these findings support the use of a shared modeling strategy rather than training a separate model for each watershed. Therefore, we adopt a shared model in the subsequent experiments.

After establishing the shared modeling strategy, we next determine the input features used in subsequent experiments by selecting the static attributes and the input forcing-window length.

Static attribute selection. Table 4 evaluates different static attribute combinations by incrementally adding watershed attributes. The results show that static attributes influence predictive performance, but their benefits vary across evaluation objectives. Among the tested combinations, {Elevation, Land} provides the strongest overall performance, particularly in terms of general predictive

Table 6

In-domain temporal predictive performance comparing the PBM and learning-based models, including the proposed MoE- τ (with 8 experts).

Model	NSE ($-\infty, 1$] $\uparrow$	MSE [0, ∞) $\downarrow$	RMSE [0, ∞) $\downarrow$	KGE ($-\infty, 1$] $\uparrow$	FHV $\rightarrow 0$	FLV $\rightarrow 0$	Pearson-r [-1, 1] $\uparrow$	Peak-MAPE [0, ∞) $\downarrow$	Peak-Timing $\rightarrow 0$
PBM	0.762	1.296	1.026	0.799	7.695	8.958	0.911	36.574	0.466
GRU	0.791	1.414	1.048	0.696	-20.508	-24.286	0.910	33.884	0.395
LSTM	0.855	<u>0.904</u>	<u>0.862</u>	<u>0.837</u>	-12.370	-53.770	<u>0.930</u>	<u>29.520</u>	<u>0.402</u>
LSTM-Attn	0.847	0.957	0.882	0.829	-9.683	-150.601	0.927	30.489	0.438
Bi-LSTM	<u>0.857</u>	0.907	0.863	0.829	-14.470	-126.312	<u>0.930</u>	30.811	0.412
CNN-LSTM	<u>0.852</u>	0.955	0.885	0.813	-18.530	24.727	0.929	33.256	0.440
MoE- τ	0.873	0.813	0.813	0.839	<u>-9.475</u>	<u>-22.226</u>	0.939	28.513	0.422

Table 7

Performance comparison between PBM and MoE- τ (with 8 experts) under a reversed train-test split, where the training period is from 1993/01/01 to 2018/12/31 and the testing period is from 1987/10/01 to 1992/12/31.

Data Split 2	NSE ($-\infty, 1$] $\uparrow$	MSE [0, ∞) $\downarrow$	RMSE [0, ∞) $\downarrow$	KGE ($-\infty, 1$] $\uparrow$	FHV $\rightarrow 0$	FLV $\rightarrow 0$	Pearson-r [-1, 1] $\uparrow$	Peak-MAPE [0, ∞) $\downarrow$	Peak-Timing $\rightarrow 0$
PBM	-0.631	0.776	0.791	0.412	28.975	46.405	0.681	57.150	0.697
MoE- τ	0.689	0.451	0.520	0.666	7.760	-17.227	0.891	46.789	0.559

skill and peak-flow magnitude estimation. Different attributes can also benefit specific aspects of streamflow prediction. For example, adding the soil type attribute improves the model's ability to capture the timing of peak-flow events. These experiments represent a subset of all possible static attribute combinations. Since the results show that static attributes affect predictive performance, an exhaustive evaluation of all possible combinations may be useful in practical applications to determine the most suitable static attribute set. Based on the overall results, we use {Elevation, Land} as the static attribute set in the subsequent experiments.

Input forcing-window selection. Table 5 evaluates different input forcing-window lengths for the dynamic meteorological forcings, including one month, six months, one year, and two years. The results show that increasing the input window generally improves streamflow prediction performance by providing more historical meteorological information. However, an excessively long window can degrade performance, possibly because it introduces less relevant or redundant historical information. Among the tested configurations, the one-year input forcing window achieves the best overall performance; therefore, we use this setting in the subsequent experiments.

4.3. Stage I: In-domain temporal predictive benchmark

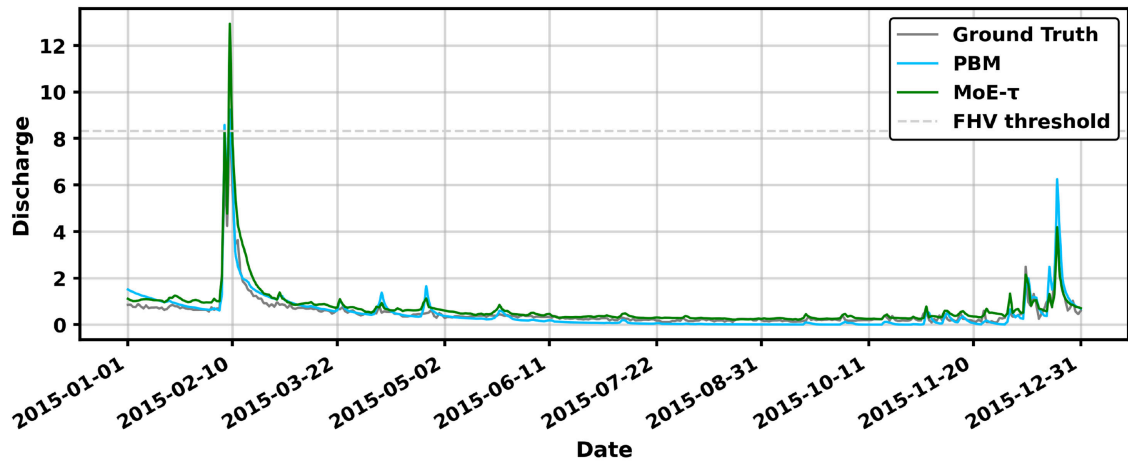
In this section, we first provide quantitative comparisons between the PBM and learning-based models under the in-domain temporal predictive benchmark, as summarized in Table 6. For each metric, we report the value averaged across all 14 watersheds. The proposed MoE- τ achieves the best performance among the learning-based models on all general-flow metrics, including NSE, MSE, RMSE, KGE, and Pearson- r , and on most extreme-flow metrics except Peak-Timing, suggesting that the mixture-of-experts design improves in-domain temporal prediction capacity.

Different learning-based architectures show metric-specific strengths. Compared with LSTM, GRU performs better on FLV and Peak-Timing, but it is less competitive on the general-flow metrics. Incorporating attention into LSTM improves high-flow behavior as measured by FHV, but leads to weaker performance on several other metrics, especially FLV. Bi-LSTM and CNN-LSTM do not show clear overall advantages in this benchmark.

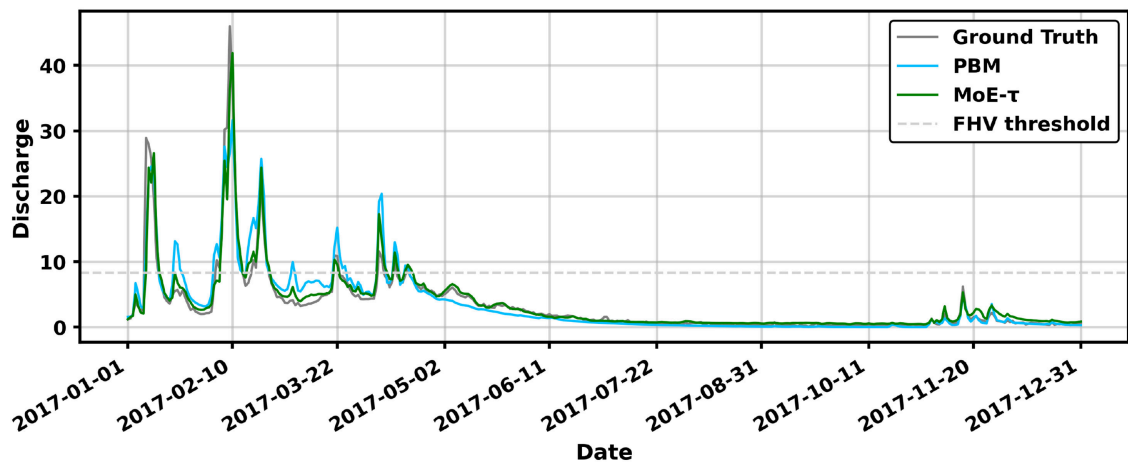
Comparing MoE- τ with PBM shows that the two approaches have different strengths. MoE- τ consistently performs better on the general-flow metrics, whereas PBM achieves better results on some extreme-flow metrics, namely FHV and FLV. For high-flow evaluation, FHV measures the signed percent bias in the total volume of the upper portion of the flow duration curve, while Peak-MAPE evaluates the absolute relative error at individual observed peak-flow events. Therefore, PBM shows stronger performance in preserving the overall high-flow volume, whereas MoE- τ better matches the magnitudes of individual peak-flow events.

These complementary behaviors are illustrated in Fig. 3, which compares the two models against observations on the Feather River at Oroville (ORO) for two hydrologically distinct testing periods. In the 2015 drought year (Fig. 3(a)), MoE- τ more accurately sustains the June–November baseflow. In the 2017 wet year (Fig. 3(b)), MoE- τ better reproduces the winter–spring peak sequence and ensuing recession limbs more faithfully, while PBM systematically overestimates the mid-size peaks and then recedes too rapidly, holding discharge below the observed values through the summer low-flow period. The two years demonstrate that MoE- τ adapts across opposite hydroclimatic extremes, capturing peak magnitude and recession dynamics in a wet year and baseflow persistence in a drought year, while the characteristic PBM biases of peak overestimation, sluggish recession, and baseflow underestimation persist across both.

To provide a more complete comparison between PBM and MoE- τ , we present watershed-wise dumbbell plots in Appendix C, Figs. C.10–C.18. These plots include all nine evaluation metrics, with each figure showing the metric values of PBM and MoE- τ across all 14 watersheds. While Table 6 reports average performance across watersheds, the dumbbell plots reveal the individual watershed-level performance patterns. The watershed-wise comparisons show that, for all evaluation metrics, MoE- τ outperforms PBM on the



(a) Critically dry water year 2015.



(b) Exceptionally wet water year 2017.

Fig. 3. Hydrograph comparison for the Feather River at Oroville (ORO). Daily discharge (runoff depth, mm/day) simulated by the PBM and the proposed MoE- τ against the observed Ground Truth over two contrasting testing-period water years: (a) 2015, a critically dry year, and (b) 2017, one of the wettest years on record. The dashed horizontal line marks the FHV (top 2% discharge) threshold. In the 2017 wet year, MoE- τ better reproduces both the winter–spring peak magnitudes and the subsequent recession limbs, whereas PBM overestimates the mid-size peaks and recedes too slowly; in the 2015 drought year, MoE- τ more accurately sustains the low June–November baseflow.

majority of watersheds, indicating that its advantage is not driven by large gains on only a few specific watersheds but is instead observed consistently across different watersheds.

We further compare MoE- τ and PBM under an alternative train–test split, as shown in Table 7. In this setting, MoE- τ outperforms PBM across all evaluation metrics. The PBM results noticeably vary between Tables 6 and 7, indicating greater sensitivity to the selected calibration period. Overall, these findings suggest that the relative advantages of learning-based models and PBMs are setting-dependent, reflecting differences in their inductive biases and their sensitivity to the selected train–test split. Therefore, ensemble methods that incorporate the strengths of different model families can be further studied.

In addition to the quantitative results, more qualitative hydrograph comparisons are provided in Appendices D and E to visualize the prediction behavior of different models. Appendix D, Figs. D.19–D.26, presents hydrographs for selected watersheds and testing periods covering both wet and dry seasons, while Appendix E, Figs. E.27–E.52, provides hydrograph comparisons between PBM and MoE- τ across all 14 watersheds for 2015 (drought year) and 2017 (wet year). These visualizations show that MoE- τ generally follows the observed streamflow trends more closely and captures both high-flow and low-flow conditions more accurately than the compared methods. Consistent with the quantitative results, the hydrographs also indicate that peak timing remains a challenging aspect for further improvement.

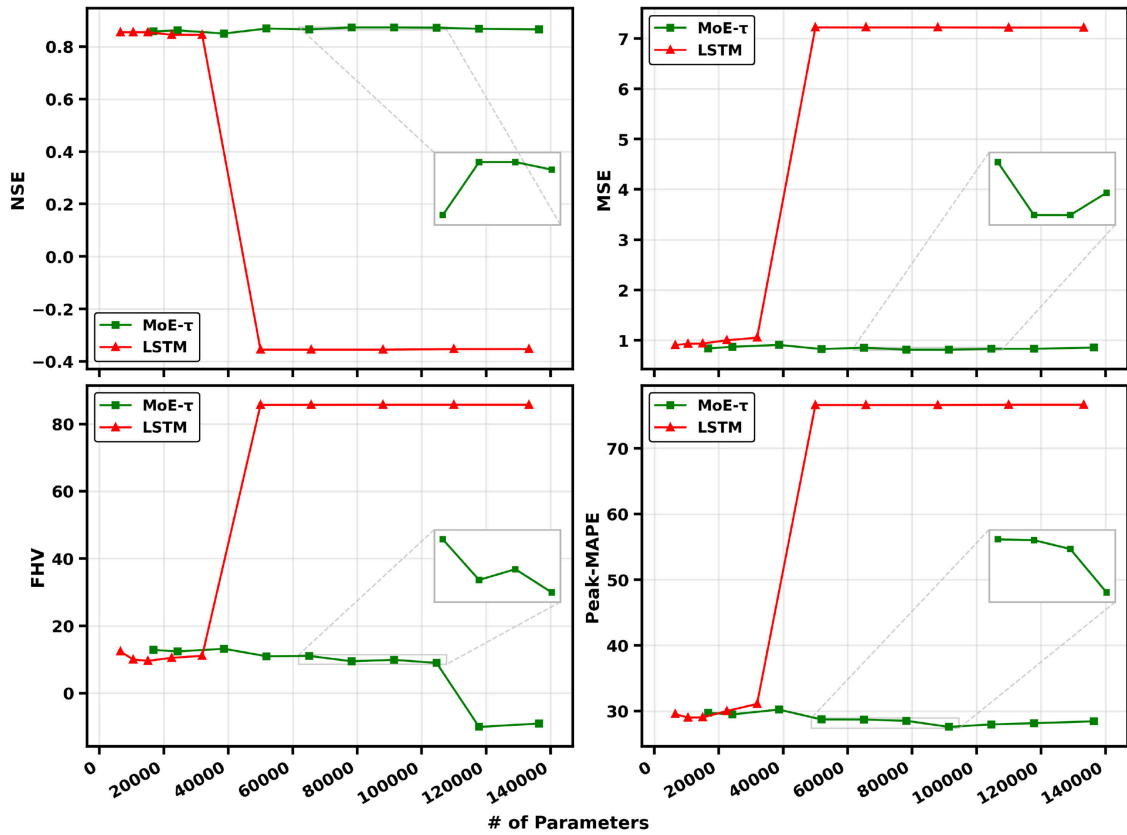


Fig. 4. MoE- τ vs. LSTM performance on four evaluation metrics (NSE, MSE, FHV, and Peak-MAPE) as model parameter size increases. The x-axis denotes the parameter size, and the y-axis shows the corresponding metric value.

Table 8

Ablation study of the proposed MoE- τ architecture. ATGN denotes the Attentive Temporal Gating Network; τ denotes the temperature in the gating softmax.

	NSE ($-\infty, 1$) $\uparrow$	MSE [0, ∞) $\downarrow$	RMSE [0, ∞) $\downarrow$	KGE ($-\infty, 1$) $\uparrow$	FHV $\rightarrow 0$	FLV $\rightarrow 0$	Pearson-r [-1, 1] $\uparrow$	Peak-MAPE [0, ∞) $\downarrow$	Peak-Timing $\rightarrow 0$
MoE with LSTM Gate	0.856	0.896	0.856	0.831	-11.841	-76.846	0.931	31.267	0.412
MoE with ATGN	0.866	0.825	0.825	0.831	-13.105	-71.075	0.935	29.270	0.434
MoE- τ ($\tau = 0.3$) with ATGN	0.871	0.821	0.813	0.830	-10.857	-47.711	0.939	30.212	0.406
MoE- τ (τ learnable) with ATGN	0.873	0.813	0.813	0.839	-9.475	-22.226	0.939	28.513	0.422

4.4. Analysis and interpretation of MoE- τ

Parameter-scale analysis. MoE- τ contains multiple LSTM experts, raising the question of whether its improvement mainly benefits from having more LSTM units. To examine this, Fig. 4 compares LSTM and MoE- τ under gradually increasing parameter scales. For LSTM, the parameter size is increased by enlarging the model size, such as stacking additional LSTM layers, while for MoE- τ , the parameter size is increased by introducing more LSTM experts. The results show that simply increasing the parameter scale of LSTM does not achieve the performance obtained by MoE- τ . In particular, larger LSTM models still fail to match MoE- τ and may suffer from degraded performance due to overfitting. This suggests that the advantage of MoE- τ comes not only from increased parameter size, but also from its expert-based modeling design.

Component ablation. We conduct an ablation study to evaluate the contribution of each component in the proposed MoE- τ architecture, as summarized in Table 8. Since the gating network determines the sequence-level contribution of each LSTM expert, its ability to extract informative temporal context is critical for effective expert weighting. We first replace the LSTM gate with the proposed Attentive Temporal Gating Network (ATGN), which applies temporal attention over the gating LSTM hidden states to form a context representation for expert selection. Comparing rows 1 and 2, ATGN improves most evaluation metrics, suggesting that attention-based temporal aggregation provides more informative gating signals than using the LSTM gate alone. In row 3, we further introduce a temperature parameter τ into the gating softmax to control the sharpness of the expert-weight distribution. The fixed value $\tau = 0.3$ is selected through multiple experiments with different temperature values and leads to additional performance gains.

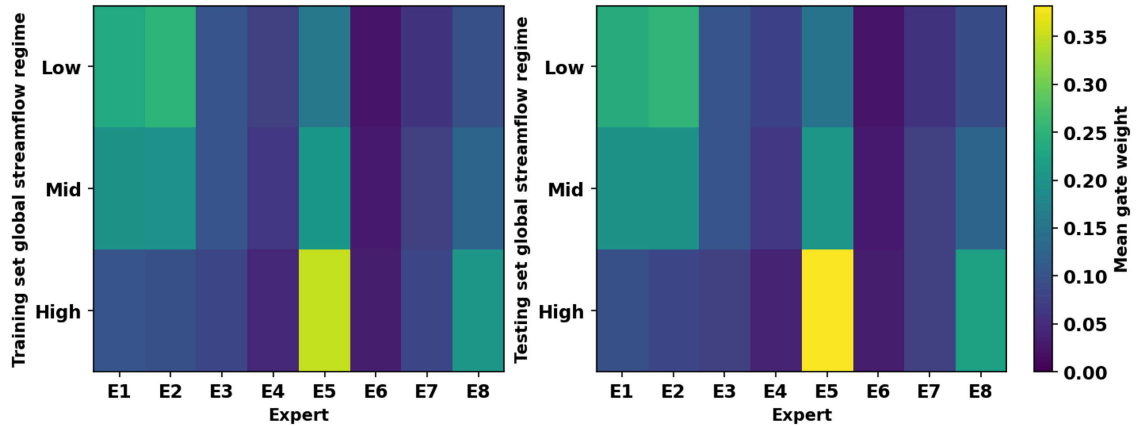


Fig. 5. Heatmaps show the average gating weights assigned to each expert under low-, mid-, and high-flow regimes for the training and testing sets.

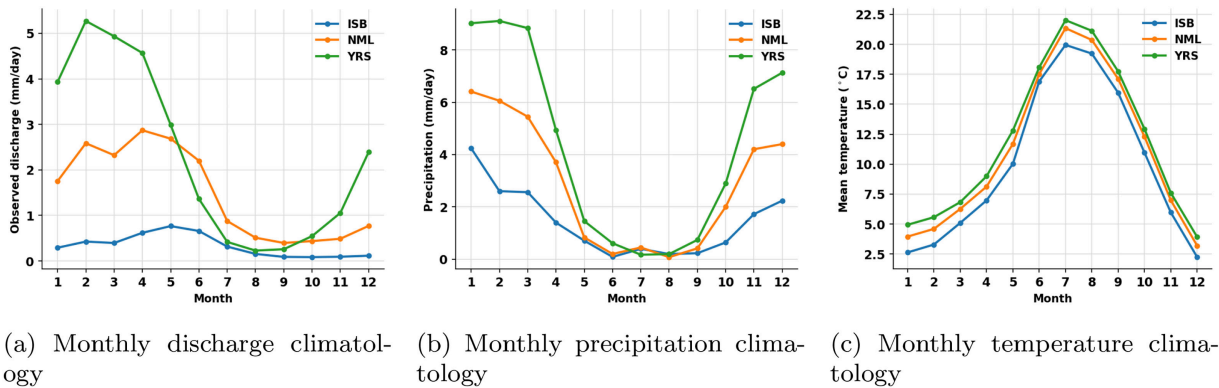


Fig. 6. Hydrometeorological distribution differences among representative watersheds. The figure compares the monthly climatology of discharge, precipitation, and temperature for three representative watersheds selected from the three hydrologic regions in this study.

However, manually tuning τ increases computational cost and may not be optimal across different settings. Therefore, in row 4, we make τ learnable and optimize it jointly with the model parameters, which achieves the best overall performance without requiring exhaustive temperature search.

Expert specialization across flow regimes. To interpret the learned gating behavior of MoE- τ , we analyze the expert weights produced by its gating network, as shown in Fig. 5. For each input sample, the gating network outputs a set of softmax weights indicating the contribution of each expert to the final prediction. We group samples into low-, mid-, and high-flow regimes using thresholds derived from the pooled training discharge distribution across all watersheds. Low-flow samples are defined as discharge values below the 30th percentile, following the FLV metric definition, while high-flow samples are defined as discharge values above the 98th percentile, following the FHV metric definition. The remaining samples are assigned to the mid-flow regime. The same training-derived thresholds are then applied to the testing set. For each flow regime, we average the gating weights over all samples in that regime and visualize the resulting expert-usage patterns as heatmaps.

The testing set heatmap shows clear regime-dependent expert specialization. Experts 5 and 8 are more strongly activated for high-flow samples, Experts 1, 2, and 5 contribute more to mid-flow samples, and Experts 1 and 2 dominate low-flow samples. The training set heatmap exhibits a similar activation pattern, indicating that MoE- τ learns consistent expert usage across training and testing periods. Some experts are activated less frequently, which may suggest that they capture less dominant hydrologic patterns or that fewer experts may be sufficient for this dataset.

Watershed distribution and expert activation. Having observed that different experts specialize in different flow regimes, we next examine how expert activations vary across individual watersheds with distinct hydrometeorological distributions. To illustrate this relationship, we select three representative watersheds from the three hydrologic regions in this study, which are ISB, NML, and YRS. These watersheds are representative in the sense that they exhibit distinct data distributions, as shown in Fig. 6. The figure compares their monthly climatology of discharge, precipitation, and temperature, where each monthly value is computed by averaging samples from the same calendar month over the entire dataset. The differences are most evident in discharge and precipitation: ISB represents relatively dry streamflow conditions, NML represents intermediate conditions, and YRS represents wetter conditions.

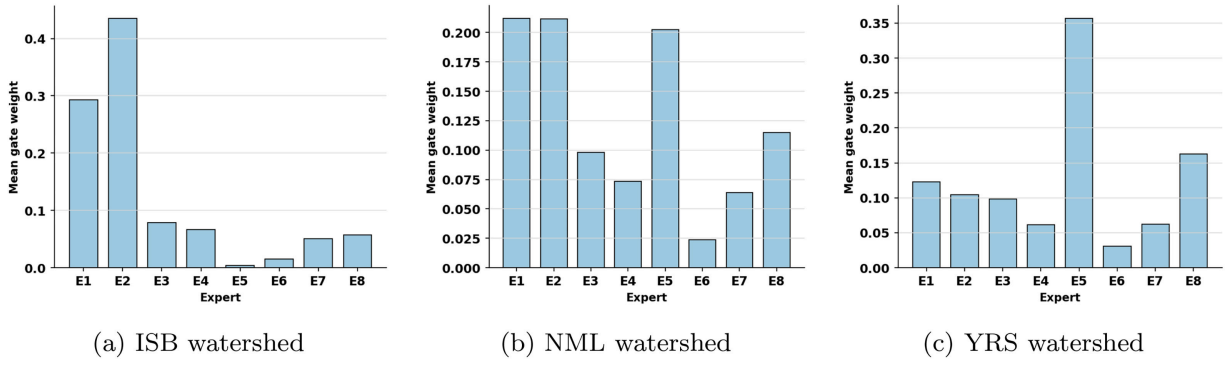


Fig. 7. Expert usage distributions of three representative watersheds.

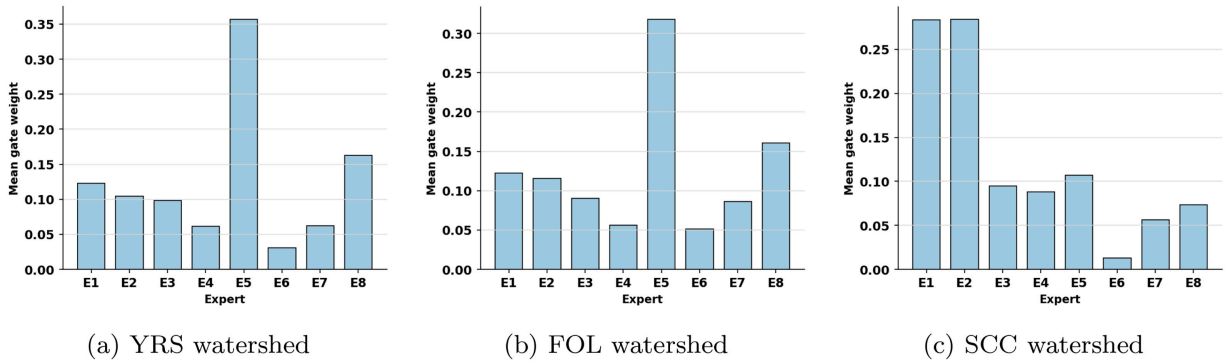


Fig. 8. Expert usage distributions of hydrometeorologically similar and different watersheds.

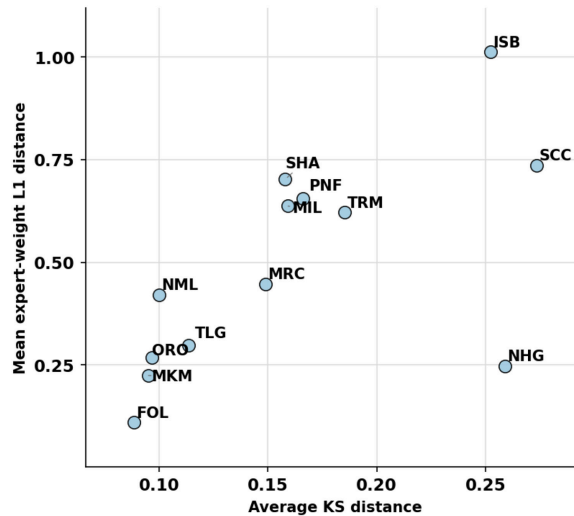


Fig. 9. Relationship between hydrometeorological distribution distance and expert-usage distance.

We then compare the corresponding expert activations of these three watersheds in Fig. 7. For each watershed, the bar plot reports the mean gating weight of each expert over all testing samples. The three watersheds exhibit different expert-usage profiles. The relatively drier ISB watershed shows stronger activation of Experts 1 and 2, the intermediate NML watershed has a more uniform expert-usage distribution, and the wetter YRS watershed shows stronger activation of Experts 5 and 8. This watershed-level pattern is consistent with the global expert-specialization trend in Fig. 5, further indicating that different hydrometeorological distributions lead MoE- τ to activate different combinations of experts.

To complement the observation, we next ask whether watersheds with similar distributions also activate similar experts. Using YRS as the reference watershed, we measure its distributional difference from each other watershed using the Kolmogorov–Smirnov

Table 9

Held-out watershed testing performance comparing LSTM with the proposed MoE- τ (with 8 experts). For each listed watershed, each model was trained on the remaining 13 watersheds and evaluated on the held-out watershed.

	Model	NSE ($-\infty, 1$) $\uparrow$	MSE [0, ∞) $\downarrow$	RMSE [0, ∞) $\downarrow$	KGE ($-\infty, 1$) $\uparrow$	FHV $\rightarrow 0$	FLV $\rightarrow 0$	Pearson- r [-1, 1] $\uparrow$	Peak-MAPE [0, ∞) $\downarrow$	Peak-Timing $\rightarrow 0$
ISB	LSTM	0.189	0.230	0.478	0.063	19.764	43.225	0.815	45.492	0.927
	MoE- τ	0.541	0.130	0.360	0.466	1.727	8.762	0.827	44.164	0.923
NML	LSTM	0.814	0.974	0.985	0.876	-4.426	24.494	0.908	39.204	0.524
	MoE- τ	0.827	0.907	0.948	0.885	-0.047	7.657	0.921	35.940	0.538
YRS	LSTM	0.093	18.954	4.327	0.020	-30.209	63.416	0.715	65.280	0.674
	MoE- τ	0.515	10.143	3.162	0.456	-21.800	46.865	0.785	52.053	0.642

(KS) distance (An, 1933; Smirnov, 1939). For each variable v , including observed discharge, precipitation, and temperature, the KS distance is computed as

$$D_v(\text{YRS}, w) = \sup_x |F_{\text{YRS},v}(x) - F_{w,v}(x)|, \quad (16)$$

where $F_{\text{YRS},v}$ and $F_{w,v}$ are the empirical cumulative distribution functions of variable v for YRS and watershed w , respectively. The variable-specific distances are then averaged to obtain one overall hydrometeorological distribution distance for each watershed pair. Based on this distance, FOL is identified as the watershed most similar to YRS, whereas SCC is identified as the watershed most dissimilar to YRS. Fig. 8 compares the expert-usage distributions of YRS, FOL, and SCC. The results show that FOL has an expert-usage profile similar to YRS, while SCC exhibits a clearly different profile, indicating that similarity in hydrometeorological distributions is reflected in similarity of expert activation patterns.

To quantify this relationship across all watersheds, Fig. 9 plots the hydrometeorological distribution distance against the expert-usage distribution distance using YRS as the reference watershed. The x-axis shows the KS-based distribution distance between YRS and each candidate watershed, while the y-axis shows the L_1 distance between their mean expert-usage distributions. Each point represents one watershed. To measure the association between these two distances, we compute Spearman's rank correlation coefficient (Spearman's r) (Spearman, 1961), which evaluates the strength of a monotonic relationship between two variables based on their ranks. The resulting Spearman's r is 0.687, indicating a strong positive relationship: watersheds with more different hydrometeorological distributions tend to have more different expert-usage patterns.

4.5. Stage II: Held-out watershed generalization

After showing the in-domain temporal prediction advantage of MoE- τ , we examine whether its expert-based modeling design also improves generalization to unseen watersheds. Table 9 compares LSTM and MoE- τ under held-out watershed experiments using ISB, NML, and YRS as target watersheds. For each experiment, the target watershed is excluded from training, and the model is trained on the remaining 13 watersheds and evaluated on the held-out watershed. The results show that MoE- τ outperforms LSTM on almost all evaluation metrics across the three held-out settings. Since these target watersheds represent distinct hydrologic conditions, the results indicate that MoE- τ provides stronger spatial generalization across diverse watershed characteristics, with improvements observed for both general-flow and extreme-flow evaluation metrics.

5. Discussion

This study presents a comprehensive evaluation of recurrent neural network architectures, specifically a novel Mixture-of-Experts with Attentive Temporal Gating Network and temperature scaling (MoE- τ), for daily streamflow prediction across 14 Sierra Nevada watersheds. By comparing this approach against standard deep learning baselines (GRU, LSTM, LSTM-Attn, Bi-LSTM, CNN-LSTM) and a traditional PBM (i.e., SAC-SMA), we identify key insights regarding model generalization, physical consistency, and operational viability.

5.1. Scientific significance and novelty

The novelty and scientific significance of MoE- τ should be distinguished from the novelty of its individual neural-network elements. LSTMs, temporal attention, expert gating, and temperature-scaled Softmax are established methods. This study makes three linked contributions. First, it formulates a sequence-level regional MoE in which the temporal context of the full meteorological input window determines the weights assigned to LSTM component representations. This differs from streamflow MoE approaches based on time step-level gating or Informer experts (Rong et al., 2025; Yang & Chen, 2026). Second, it tests the hydrologic hypothesis that adaptive component weighting can represent differences among flow regimes and watershed conditions more effectively than a single shared recurrent representation. Third, it connects the learned weighting behavior to hydrologic variation and evaluates whether that behavior supports transfer to watersheds excluded from training. Learnable temperature scaling is used to adapt the concentration of the component weights during training rather than requiring a fixed value to be selected for each application.

To our knowledge, this is the first systematic application of a sequence-level, temperature-controlled LSTM Mixture-of-Experts framework to regional daily streamflow prediction across the Sierra Nevada. This application is hydrologically significant because the study watersheds span contrasting rain- and snow-influenced regimes and pronounced wet-dry differences within a water-supply region where both floods and droughts affect management. This provides a demanding setting for evaluating whether adaptive component selection can accommodate this heterogeneity, and the temporal benchmark establishes the predictive relevance of the framework: MoE- τ achieves the strongest general-flow performance among the learning-based models and the lowest peak-magnitude error, while its comparison with SAC-SMA reveals complementary strengths in aggregate extreme-flow volumes (Table 6). Thus, the contribution to hydrologic research lies in both the regional application and the evidence about when and why adaptive specialization is useful.

The architecture analyses substantiate that claim. Increasing the size of a conventional LSTM does not reproduce the performance of MoE- τ and can lead to degradation at larger parameter scales, indicating that capacity alone is insufficient to explain the gain (Fig. 4). The ablation study then isolates the contribution of the proposed design. Replacing the conventional LSTM gate with attentive temporal gating improves most metrics, temperature scaling adds further gains, and learning the temperature produces the strongest overall configuration, increasing mean NSE from 0.856 to 0.873 (Table 8). Together, Fig. 4 and Table 8 attribute the improvement to how the model selects and combines its components rather than simply to the presence of additional recurrent parameters. The component-weight analyses provide hydrologic evidence for this mechanism. Experts 1 and 2 contribute most during low flows, whereas experts 5 and 8 receive greater weight during high flows. The similarity of these patterns in the training and testing periods indicates consistent flow-regime specialization (Fig. 5). The contrasting discharge, precipitation, and temperature climatologies of representative dry, intermediate, and wet watersheds provide the hydrometeorological context for the watershed-level analysis (Fig. 6). These watersheds activate different component combinations (Fig. 7), while watersheds with similar distributions exhibit more similar component-use profiles than hydrometeorologically dissimilar watersheds (Fig. 8). Across the watershed comparisons, hydrometeorological distribution distance is positively associated with component-use distance (Spearman's $r = 0.687$; Fig. 9). These results do not prove that individual components represent unique physical processes, but they show that the weighting mechanism responds systematically to hydrologically relevant variation.

Finally, the held-out-watershed experiments test whether the proposed specialization is useful beyond the training basins. Relative to a shared LSTM, MoE- τ improves NSE by 0.352, 0.013, and 0.422 for ISB, NML, and YRS, respectively, and performs better on nearly all reported metrics across the three contrasting targets (Table 9). The larger gains for ISB and YRS, where the shared LSTM transfers poorly, are consistent with the benefit of adaptive specialization under stronger distributional differences. Because only three watersheds were held out, these results support spatial transfer within the study region but do not establish general performance for all ungauged basins.

Taken together, Tables 6, 8 and 9, Figs. 4–9 provide a connected line of evidence from predictive benefit, to architectural attribution, to hydrologic interpretation, and finally to spatial transfer. Accordingly, the main advance is not the invention of attention, LSTM, or temperature scaling individually. It is an empirically supported regional modeling strategy that uses sequence-dependent component weighting to accommodate Sierra Nevada watershed heterogeneity while retaining a measurable degree of interpretability.

5.2. Practical implications

From an operational perspective, the MoE- τ model serves as a promising tool that can supplement the existing operational process-based models in supporting critical planning studies and real-time management. Reliable daily streamflow prediction in the Sierra Nevada is particularly vital because these watersheds act as the headwaters for the SWP and CVP, which transport water to drier agricultural and urban centers. By providing more accurate predictions, the model helps enhance water supply reliability, allowing managers to better anticipate inflow volumes for allocation decisions. This is crucial for balancing the competing demands of storing water for the dry season—when agricultural and municipal needs are highest—against the necessity of maintaining flood control space in reservoirs. The model's superior performance in handling hydrologic extremes further strengthens its value for flood and drought management. Its improved accuracy in capturing peak flow magnitudes and low-flow periods supports more accurate decision-making during volatile conditions. For instance, accurate high-flow predictions enable optimized reservoir releases to mitigate flood risks, while reliable low-flow forecasts are essential for managing curtailments and preserving storage during prolonged droughts. Furthermore, the model's demonstrated ability to generalize to unseen watersheds suggests it can be effectively deployed to ungauged basins, expanding the state's capacity to monitor water resources across a spatially diverse and complex hydroclimatic network. Finally, the MoE- τ architecture offers significant advantages in computational efficiency and ease of maintenance compared to traditional methods. While process-based models often require extensive, site-specific calibration and expert intervention to replicate watershed behaviors, this data-driven approach automates feature extraction and adapts to new data with minimal manual tuning. This efficiency not only reduces the resource burden on operations but also facilitates rapid model updating, ensuring that forecasts remain robust as hydrologic conditions evolve.

To fully leverage these computational and predictive advantages, the MoE- τ architecture offers a realistic deployment path within established state and federal frameworks. For real-time operations, the model could be integrated as a supplementary tool alongside the California Nevada River Forecast Center (CNRFC)'s operational flood and water supply forecasting model SAC-SMA. Rather than replacing these deeply embedded continuous-simulation models, the MoE- τ can act as a high-speed “shadow” model or an independent ensemble member. This provides forecasters with a rapid, data-driven secondary projection to build confidence during rapidly evolving atmospheric river events or complex snowmelt dynamics. On the long-term planning side, the MoE- τ framework can

serve as an efficient pipeline to feed hydrological inputs into system-wide water planning models like CalSim (Draper et al., 2004). Generating updated inflow time series for CalSim—particularly when evaluating numerous climate change projections or shifting land-use scenarios—traditionally requires labor-intensive, site-specific calibration of process-based models. Because this data-driven approach automates feature extraction and adapts to new data with minimal manual tuning, it enables the rapid generation of these scenario-based streamflow inputs. This streamlined integration would significantly reduce the resource burden on operations, facilitating more agile updates to SWP and CVP reliability studies and long-term allocation planning.

5.3. Limitations

Despite the strong predictive performance demonstrated in this study, the MoE- τ architecture exhibits notable limitations that warrant consideration. While the model accurately captured the magnitude of peak flows, precisely replicating the timing of these events remains a challenge. As indicated by the Peak-Timing metric, the MoE- τ model occasionally predicts peaks slightly earlier or later than observed, a discrepancy that could pose challenges for real-time flood operations where the exact timing of inflows is critical for reservoir management. Furthermore, issues regarding physical consistency persist. Although the MoE- τ generally outperforms the PBM across most statistical metrics, data-driven approaches inherently lack explicit physical constraints, such as the conservation of mass. The PBM retained a slight advantage in certain high-flow volume (FHV) comparisons within specific data splits, suggesting that explicit representations of hydrological processes may still offer superior robustness in extreme extrapolation scenarios where pure machine learning models may struggle. Finally, the model's performance is heavily contingent on the quality and extent of the training data. While our experiments identified a one-year lookback window as optimal for this dataset, the architecture relies entirely on consistent meteorological forcings. Consequently, operational accuracy is susceptible to errors in input data, particularly if real-time meteorological forecasts deviate from observed conditions.

MoE- τ also introduces engineering costs relative to a single recurrent model. The implementation used here is a dense mixture: all eight LSTM experts process every input sequence, after which the attentive gate combines their representations. The expert-related parameter count, activation memory, and recurrent computation therefore increase approximately with the number of experts, with additional cost from the gating network. Although the experts can be evaluated in parallel on appropriate hardware, this parallelism does not eliminate the total computational or memory burden, and actual latency depends on hardware, batch size, input length, and software implementation. The parameter-scale analysis shows that increased capacity alone does not explain the predictive improvement, but it should not be interpreted as evidence of computational efficiency.

The computational trade-off differs markedly between training and inference. Once MoE- τ has been trained, generating predictions is fast and was completed within seconds in our implementation, supporting practical use of the trained model for rapid prediction. Training, however, is substantially more demanding than for a single LSTM: optimizing all eight experts and the gating network requires considerably greater computation time, memory, and hardware resources. Computational feasibility was not comprehensively benchmarked in this study, and standardized measurements of wall-clock training time, peak memory, energy consumption, and per-sample inference latency were not collected. Accordingly, the observed seconds-scale inference should not be interpreted as a hardware-independent speed comparison or as evidence that MoE- τ is computationally cheaper than SAC-SMA or the recurrent baselines. Future work should report hardware-controlled cost-accuracy comparisons and examine whether similar performance can be retained with fewer components, sparse top-k routing, shared parameters, pruning, or knowledge distillation. Until such tests are completed, MoE- τ should be viewed as offering a demonstrated predictive benefit and rapid inference after training, at the cost of substantially more resource- and time-intensive training than a single LSTM.

5.4. Future work

To address the current limitations and advance the state of the art in hydrological forecasting, future research should focus on bridging the gap between data-driven accuracy and physical interpretability. A primary avenue for improvement is the integration of physics-informed machine learning. By embedding physical constraints, such as mass balance loss functions, directly into the MoE- τ training process, the model could achieve higher physical consistency and potentially mitigate the peak timing errors observed in this study. Such hybrid approaches ensure that while the model learns from empirical data, its outputs remain within the bounds of known hydrological laws. Additionally, the development of hybrid modeling ensembles represents a promising direction. Given that PBMs and MoE- τ architectures exhibit complementary strengths—PBMs in physical robustness during extreme extrapolation and MoE- τ in general statistical accuracy—a framework that dynamically weights their outputs could yield a more resilient forecasting system. This would be particularly valuable for long-term water supply planning under climate change. Future work should also include stress-testing the model against synthetic future climate scenarios to evaluate its reliability under non-stationary conditions, such as warming-induced shifts from snow to rain, which are expected to significantly alter the streamflow patterns of the Sierra Nevada watersheds.

6. Conclusion

This study presented MoE- τ , a mixture-of-experts framework with an attention-based gating network and a learnable temperature mechanism for regional daily streamflow prediction across heterogeneous Sierra Nevada watersheds. The experiments lead to three main conclusions.

First, analyses supported the use of a shared watershed modeling strategy and identified elevation, land use, and a one-year input forcing window as effective input configurations. This setup provides the basis for training one regional model rather than separate models for individual watersheds.

Second, MoE- τ achieved the strongest overall in-domain temporal prediction performance among the learning-based models and showed competitive advantages over the process-based SAC-SMA model, particularly on general-flow metrics. Additional analyses indicate that these improvements are not simply due to increased parameter size. Instead, the expert-based design enables different experts to specialize in different flow regimes, and expert activation patterns are related to watershed-specific hydrometeorological distributions.

Third, the held-out watershed experiments showed that MoE- τ generalizes better than LSTM to unseen watersheds across dry, intermediate, and wet hydrologic conditions. This suggests that adaptive expert selection is useful not only for improving temporal prediction within observed watersheds, but also for supporting regional transfer to watersheds excluded from training.

Overall, the findings indicate that MoE- τ provides a flexible and interpretable learning-based approach for regional streamflow prediction in heterogeneous mountain watersheds. The model may be useful for regional streamflow prediction in data-limited or ungauged watersheds and offers a promising learning-based alternative to conventional process-based approaches. Future work should further explore hybrid process-based and learning-based ensembles, incorporate hydrologic constraints, and evaluate robustness under forecast forcings and non-stationary climate conditions.

CRediT authorship contribution statement

Yu-Chieh Chao: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization; **Wyatt Arnold:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization; **Minxue He:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization; **Zhaojun Bai:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization; **Zhi Ding:** Writing – review & editing, Supervision, Conceptualization; **Prabhjot Sandhu:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Conceptualization; **Peyman Namadi:** Writing – review & editing, Methodology; **Yiwei Cheng:** Writing – review & editing, Writing – original draft, Methodology; **Jamie Anderson:** Writing – review & editing, Methodology; **Shima Shamkhali Chenar:** Writing – review & editing, Methodology; **Elissa Lynn:** Writing – review & editing, Methodology; **K. A. Naditha Macaluso:** Writing – review & editing, Methodology; **Alejandro Perez:** Writing – review & editing, Methodology; **Andrew Schwarz:** Writing – review & editing, Methodology.

Declaration of generative AI use

During the preparation of this manuscript, the authors used generative AI tools only to improve grammar, sentence fluency, and readability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the California Department of Water Resources (Award No. 4600014977). The authors gratefully acknowledge the support of colleagues Erik Reyes, Nazrul Islam, and Darren Bonfantine throughout this work.

Data availability

The data and code were maintained in a GitHub repository and are available from the lead author upon reasonable request.

Appendix A. Detailed model configurations

All models were trained for 200 epochs with a batch size of 256 using the AdamW optimizer and MSE loss. Each model was trained 5 times using different random seeds, and the reported performance was averaged across the five runs. The initial learning rate was set to 0.01, and a ReduceLROnPlateau scheduler was applied to reduce the learning rate by a factor of 10 when the training loss did not decrease for 5 consecutive epochs. Gradient clipping with a threshold of 1.0 was used to improve training stability. The hidden dimension of the backbone models was set to 32, and the dropout rate was set to 0.4, where dropout was applied to the output representation of the backbone model, such as the LSTM output, before the final prediction layer. The input forcing window denotes the number of daily meteorological forcing records used as dynamic input, including the prediction day. During training, the dynamic forcings, static attributes, and target discharge were standardized using z-score normalization based solely on the training-set statistics. During testing, the same training-set means and standard deviations were used to standardize the input variables, and no statistics from the testing set were used for normalization. The predicted discharge was then transformed back to the original scale using the training set discharge statistics before computing the evaluation metrics.

Table A.10

Summary of experimental hyperparameters. These hyperparameters are shared by all learning-based models.

Hyperparameter	Value
Epochs	200
Batch size	256
Initial learning rate	0.01
Learning-rate scheduler	ReduceLROnPlateau
Optimizer	AdamW
Gradient clipping	1.0
Loss function	MSE
Dropout	0.4
Hidden dimension	32
Input forcing window	365 days

Table A.11

Summary of model architecture configurations.

Model	Architecture configuration
GRU	Single-layer GRU with hidden dimension 32.
LSTM	Single-layer LSTM with hidden dimension 32.
LSTM-Attn	Single-layer LSTM with hidden dimension 32, followed by a causal self-attention block with query, key, and value dimensions of 32.
Bi-LSTM	Single-layer bidirectional LSTM with hidden dimension 32 per direction, resulting in a 64-dimensional concatenated representation.
CNN-LSTM	Three-layer causal 1D CNN followed by a single-layer LSTM. Each convolutional layer uses 32 output channels, kernel size 5, dilation 1, batch normalization, ReLU activation, and dropout. No temporal pooling is applied. The LSTM hidden dimension is 32.
MoE- τ	Mixture-of-experts model with 8 parallel single-layer LSTM experts, each with hidden dimension 32. The gating network uses a single-layer LSTM with hidden dimension 32, temporal attention pooling, and a linear projection to produce 8 expert weights. The gating softmax uses a learnable temperature parameter τ , initialized to 0.3.

Appendix B. SAC-SMA model formulation and calibration

This appendix gives the governing equations of the four coupled modules of the SAC-SMA process-based model and lists the parameters of the calibrated model of [Wi and Steinschneider \(2022\)](#) adopted here ([Table B.12](#)). Symbol names follow the model implementation.

B.1. Hamon potential evapotranspiration

Daily potential evapotranspiration is a function of daylength and a saturation vapour pressure term, scaled by the calibrated coefficient K_{pet} :

$$E_p = K_{\text{pet}} \cdot 29.8 D \frac{e_s(T_a)}{T_a + 273.2},$$

$$e_s(T_a) = 0.611 \exp\left(\frac{17.27 T_a}{237.3 + T_a}\right),$$
(B.1)

where E_p is PET (mm day⁻¹), D the daylight length (h), T_a the mean daily air temperature (°C), and e_s the saturation vapour pressure (kPa).

B.2. Snow-17 accumulation and melt

Precipitation falls as snow when $T_a \leq \text{PXTEMP}$ and as rain otherwise. During non-rain periods, melt M follows a degree-day formulation with a seasonally varying melt factor M_f that oscillates between MFMIN and MFMAX:

$$M_f = \frac{\Delta t}{6} [S_v (\text{MFMAX} - \text{MFMIN}) + \text{MFMIN}],$$

$$M = M_f (T_a - \text{MBASE}) + 0.0125 P_r T_r,$$
(B.2)

where $S_v \in [0, 1]$ is a sinusoidal seasonal index peaking near the summer solstice, P_r is rainfall and T_r its temperature. During rain-on-snow events, melt is computed from a surface energy balance governed by the wind coefficient UADJ. The snowpack is tracked through its ice content, liquid-water content, and heat deficit, and releases P_{eff} (melt plus rain) to the SMA module once the pack is ripe.

Table B.12
SAC-SMA parameters, feasible calibration ranges, and calibration treatment.

Module	Param.	Description	Unit	Range	Calibration
PET	K_{pet}	Hamon PET proportionality coefficient	–	0.4–2.5	Distributed (veg.)
SMA	UZTWM	Upper-zone tension water capacity	mm	1–1000	Distributed (soil)
	UZFWM	Upper-zone free water capacity	mm	1–1000	Distributed (soil)
	LZTWM	Lower-zone tension water capacity	mm	50–5000	Distributed (soil)
	LZFPM	Lower-zone primary free water capacity	mm	50–5000	Distributed (soil)
	LZFSM	Lower-zone supplementary free water capacity	mm	50–5000	Distributed (soil)
	UZK	Upper-zone free-water depletion rate	day ^{−1}	0.01–0.99	Distributed (soil)
	LZPK	Lower-zone primary depletion rate	day ^{−1}	0.01–0.5	Distributed (soil)
	LZSK	Lower-zone supplementary depletion rate	day ^{−1}	0.01–0.5	Distributed (soil)
	ZPERC	Percolation demand scale	–	1–500	Distributed (soil)
	REXP	Percolation demand shape exponent	–	1–10	Distributed (soil)
	PFREE	Fraction percolating to lower-zone free water	–	0–0.99	Distributed (soil)
	PCTIM	Permanent impervious fraction	–	0–0.5	Distributed (soil)
	ADIMP	Additional (variable) impervious fraction	–	0–0.9	Distributed (soil)
	RIVA	Riparian vegetation area fraction	–	0–0.9	Distributed (soil)
	RSERV	Lower-zone free water not transferable to tension	–	0–0.4	Distributed (soil)
	SIDE	Deep-recharge-to-channel-baseflow ratio	–	0	Fixed
Snow (Snow-17)	SCF	Snowfall gauge-catch correction factor	–	1.0	Fixed
	PXTEMP	Rain/snow temperature threshold	°C	0	Fixed
	MFMAX	Maximum (summer) non-rain melt factor	–	0.05–5.0	Lumped
	MFMIN	Minimum (winter) non-rain melt factor	–	0.05–5.0	Lumped
	UADJ	Rain-on-snow wind function	–	0.03–0.2	Lumped
	MBASE	Base temperature for non-rain melt	°C	0–5	Lumped
	TIPM	Antecedent temperature index weight	–	0.1–1.0	Lumped
	PLWHC	Liquid-water holding capacity	–	0.02–0.3	Lumped
	NMF	Maximum negative melt factor	–	0.05–0.3	Lumped
	DAYGM	Daily ground melt at snow-soil interface	mm	0–0.3	Lumped
Routing (Lohmann)	N	HRU gamma unit-hydrograph shape	–	1–20	Lumped
	K	HRU gamma unit-hydrograph rate	–	0.01–0.99	Lumped
	VELO	Saint-Venant wave velocity	m s ^{−1}	0.5–5	Lumped
	DIFF	Saint-Venant diffusivity	m ² s ^{−1}	200–4000	Lumped

B.3. Sacramento soil-moisture accounting

Effective input fills an upper zone (tension and free water) and a lower zone (tension, primary, and supplementary free water). Percolation from the upper free-water store to the lower zone is demand-driven:

$$\begin{aligned}
 \text{PERC} &= \overline{P_{\text{base}}} \frac{\text{UZFWC}}{\text{UZFWM}} \left[1 + \text{ZPERC} (\text{DEFR})^{\text{REXP}} \right], \\
 \overline{P_{\text{base}}} &= \text{LZFPM} \text{LZPK} + \text{LZFSM} \text{LZSK}, \\
 \text{DEFR} &= 1 - \frac{\text{LZTWC} + \text{LZFPC} + \text{LZFSC}}{\text{LZTWM} + \text{LZFPM} + \text{LZFSM}}.
 \end{aligned} \tag{B.3}$$

The runoff components are interflow $I = \text{UZFWC} \cdot \text{UZK}$, baseflow $Q_{\text{base}} = \text{LZFPC} \cdot \text{LZPK} + \text{LZFSC} \cdot \text{LZSK}$, and impervious and surface runoff generated when the upper free-water store exceeds its capacity UZFWM; evapotranspiration is withdrawn from the tension stores at the rate set by E_p . These fluxes close the per-HRU balance, Eq. (1).

B.4. Lohmann routing

Within each HRU the unit hydrograph $u(t)$ is a gamma distribution, and the channel response $H(x, t)$ uses the Green's function of the linearized Saint-Venant equation:

$$\begin{aligned}
 u(t) &= \frac{K^N}{\Gamma(N)} t^{N-1} e^{-Kt}, \\
 H(x, t) &= \frac{x}{2t\sqrt{\pi t \text{DIFF}}} \exp \left[-\frac{(\text{VELO} t - x)^2}{4 \text{DIFF} t} \right],
 \end{aligned} \tag{B.4}$$

where N and K are the gamma shape and rate parameters, x is the channel distance to the outlet, VELO the wave velocity, and DIFF the diffusivity. Outlet discharge is the convolution of the combined unit hydrograph with the HRU runoff, summed over all HRUs.

B.5. Parameters and calibration

The 31 SAC-SMA parameters, their feasible ranges, and their distributed, lumped, or fixed treatment are listed in Table B.12. Using this parameterization, [Wi and Steinschneider \(2022\)](#) calibrated the model with the genetic algorithm of [Wang \(1991\)](#) under the pooled, multi-basin scheme of [Wi et al. \(2015\)](#), maximizing the mean KGE across all gauges over more than 6,000 HRUs. The algorithm used a population of 250, two elite members, a crossover fraction of 0.8 with scattered crossover, Gaussian mutation, and rank-based fitness scaling, evolved over 100 generations (25,000 function evaluations); the initial population was drawn uniformly within the feasible ranges of Table B.12. Calibration spanned 1989–2003 and validation 2004–2018. We adopt their calibrated parameter set without modification.

Appendix C. Pairwise performance comparison between PBM and MoE- τ

Figs. C.10–C.18.

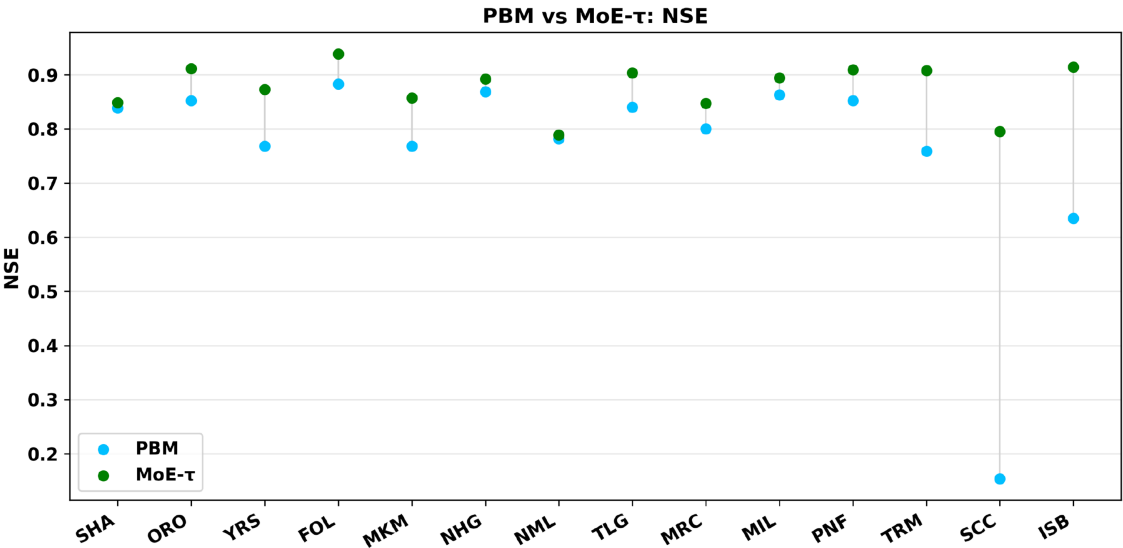


Fig. C.10. Pairwise comparison of PBM and MoE- τ across all 14 watersheds in terms of NSE.

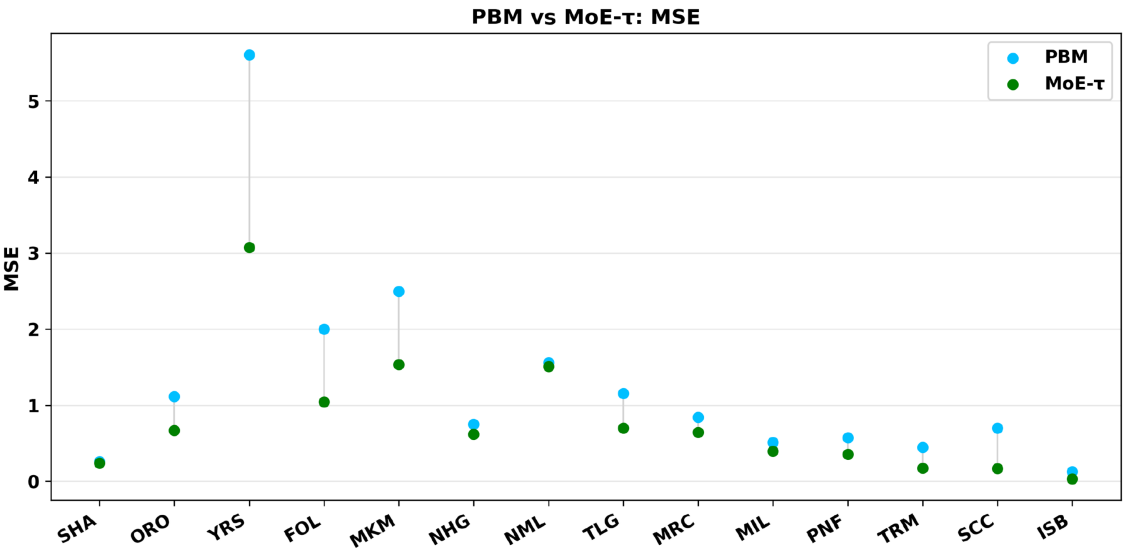


Fig. C.11. Pairwise comparison of PBM and MoE- τ across all 14 watersheds in terms of MSE.

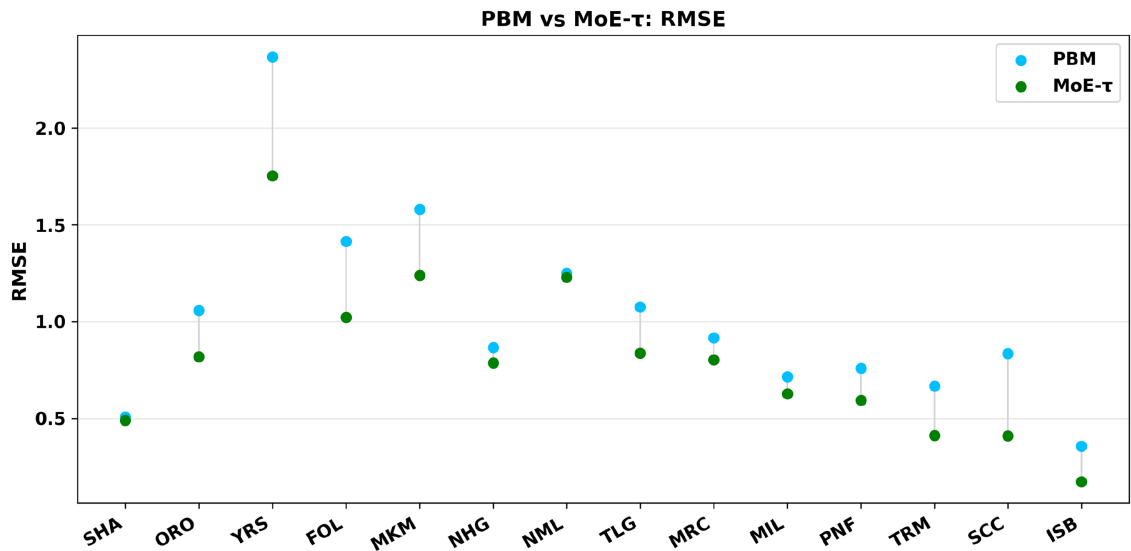


Fig. C.12. Pairwise comparison of PBM and MoE- τ across all 14 watersheds in terms of RMSE.

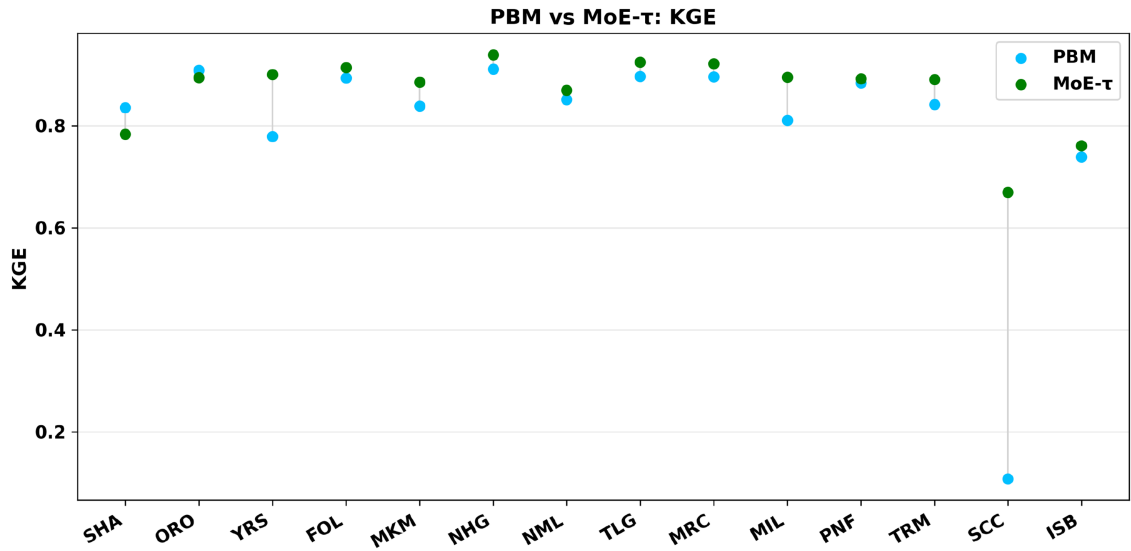


Fig. C.13. Pairwise comparison of PBM and MoE- τ across all 14 watersheds in terms of KGE.

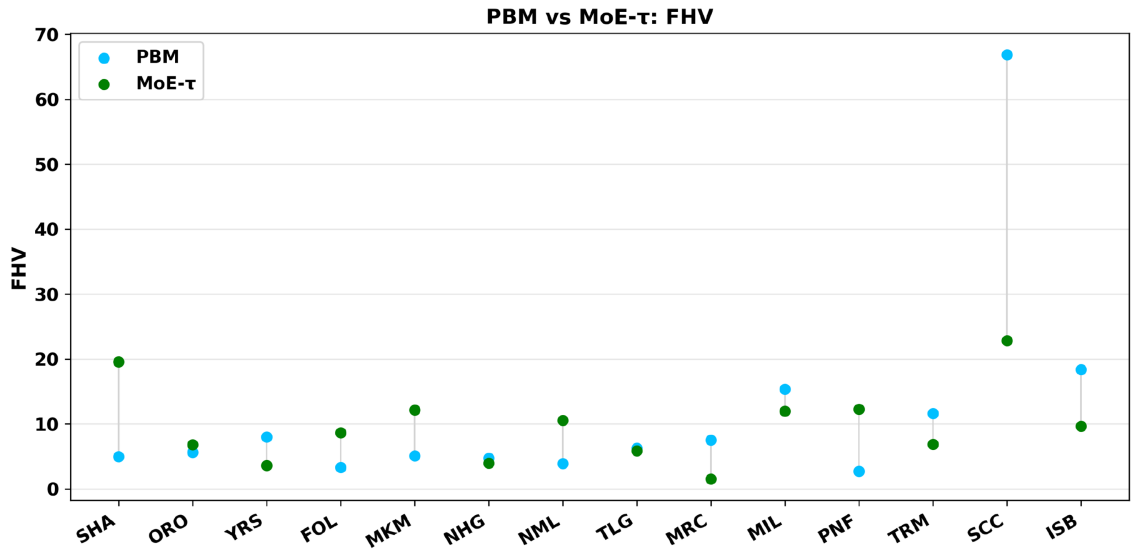


Fig. C.14. Pairwise comparison of PBM and MoE- τ across all 14 watersheds in terms of FHV. For visualization purposes, the absolute values of FHV are presented, such that lower values correspond to better performance.

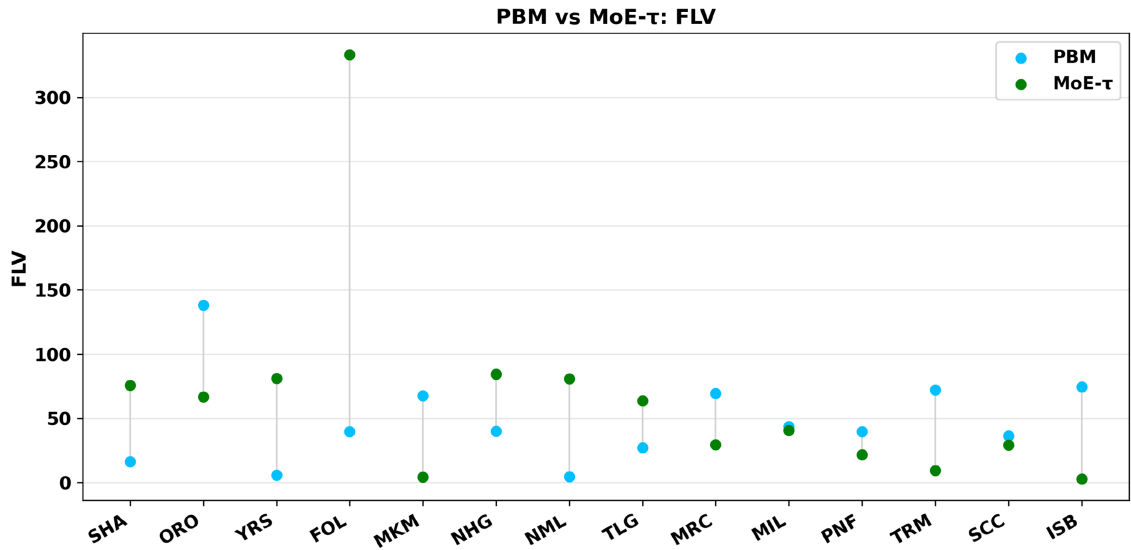


Fig. C.15. Pairwise comparison of PBM and MoE- τ across all 14 watersheds in terms of FLV. For visualization purposes, the absolute values of FLV are presented, such that lower values correspond to better performance.

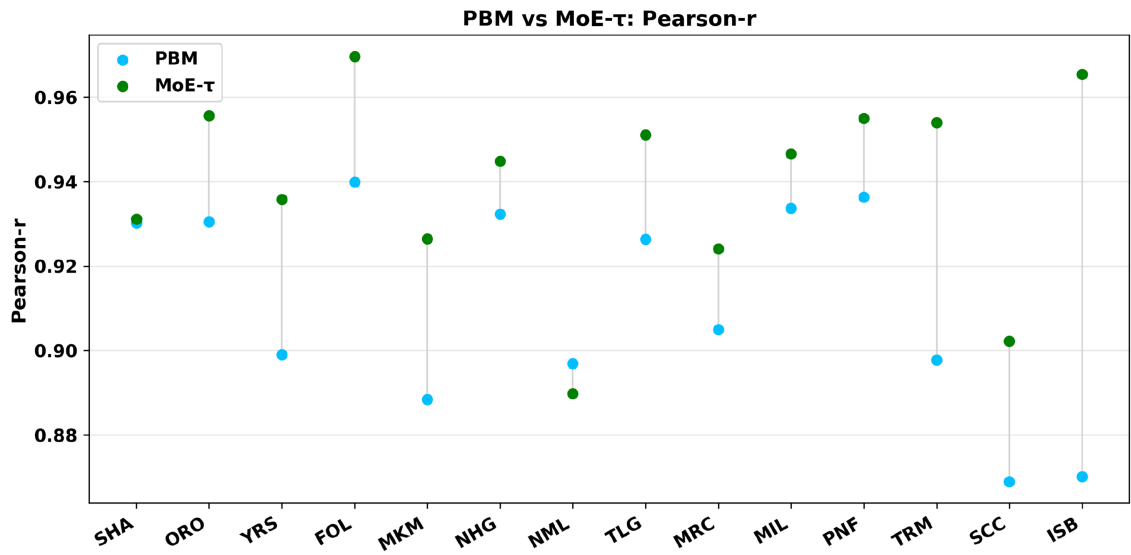


Fig. C.16. Pairwise comparison of PBM and MoE- τ across all 14 watersheds in terms of Pearson-r.

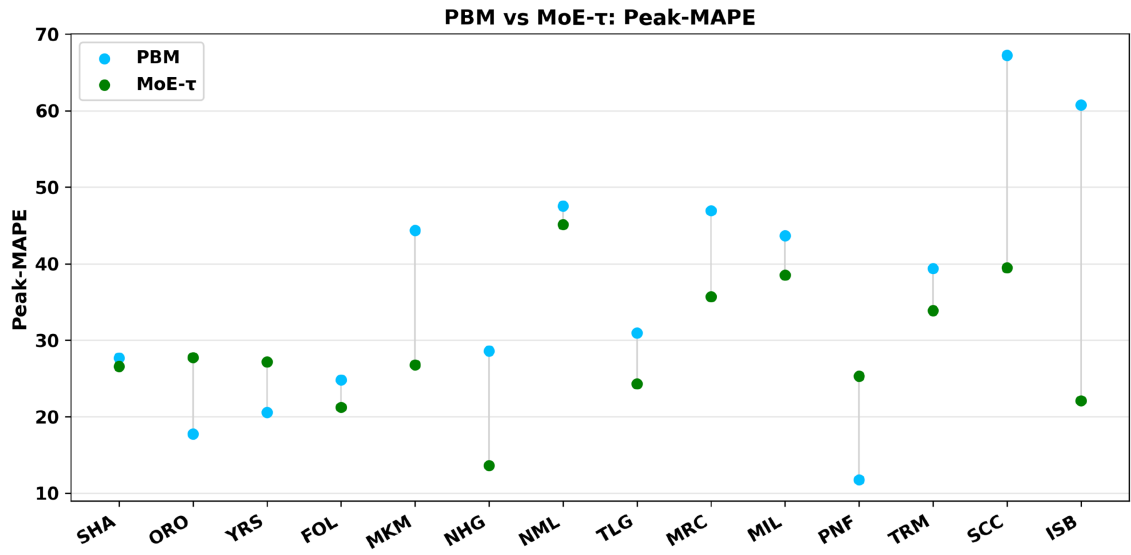


Fig. C.17. Pairwise comparison of PBM and MoE- τ across all 14 watersheds in terms of Peak-MAPE.

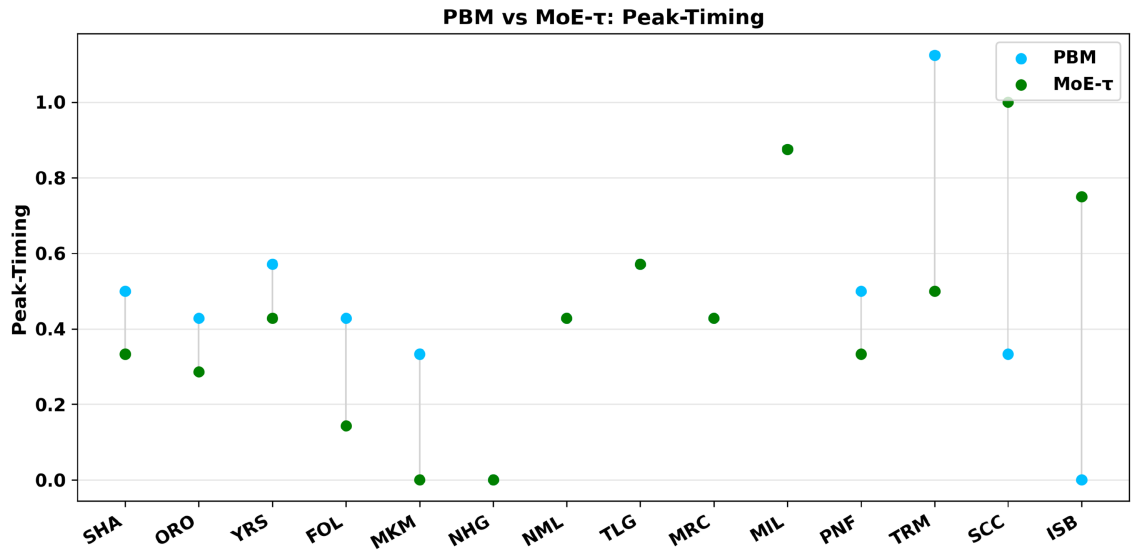


Fig. C.18. Pairwise comparison of PBM and MoE- τ across all 14 watersheds in terms of Peak-Timing. For visualization purposes, the absolute values of Peak-Timing are presented, such that lower values correspond to better performance.

Appendix D. Hydrograph comparison on selected watersheds and testing periods

Figs. D.19–D.26.

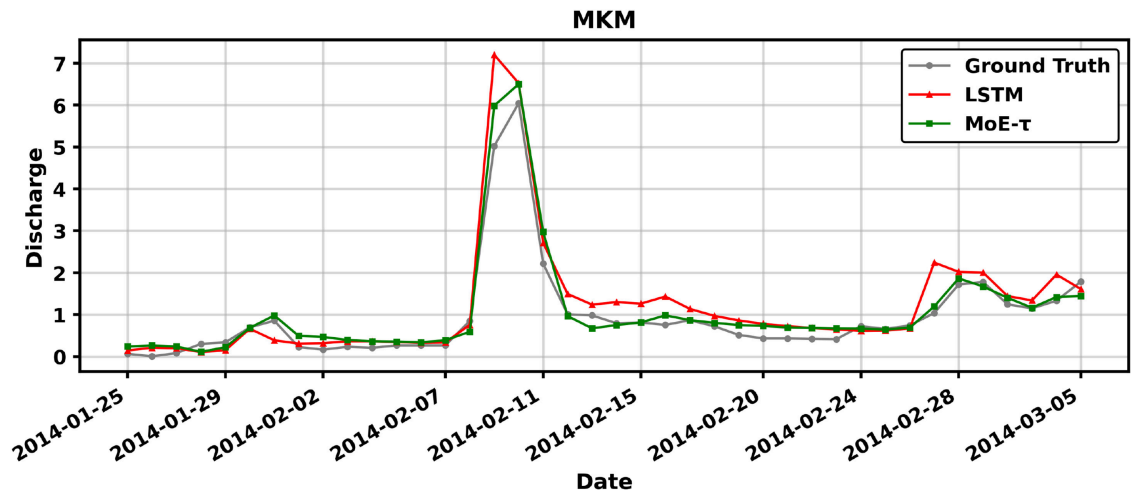


Fig. D.19. Visualization comparison of LSTM and MoE- τ discharge predictions for the MKM watershed over the period 2014/01/25 ~ 2014/03/05.

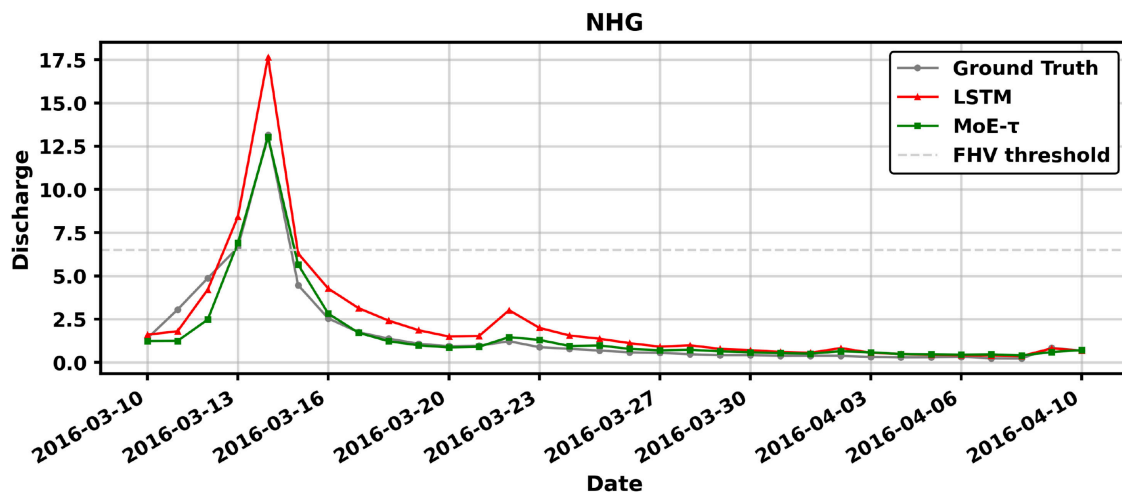


Fig. D.20. Visualization comparison of LSTM and MoE- τ discharge predictions for the NHG watershed over the period 2016/03/10 ~ 2016/04/10.

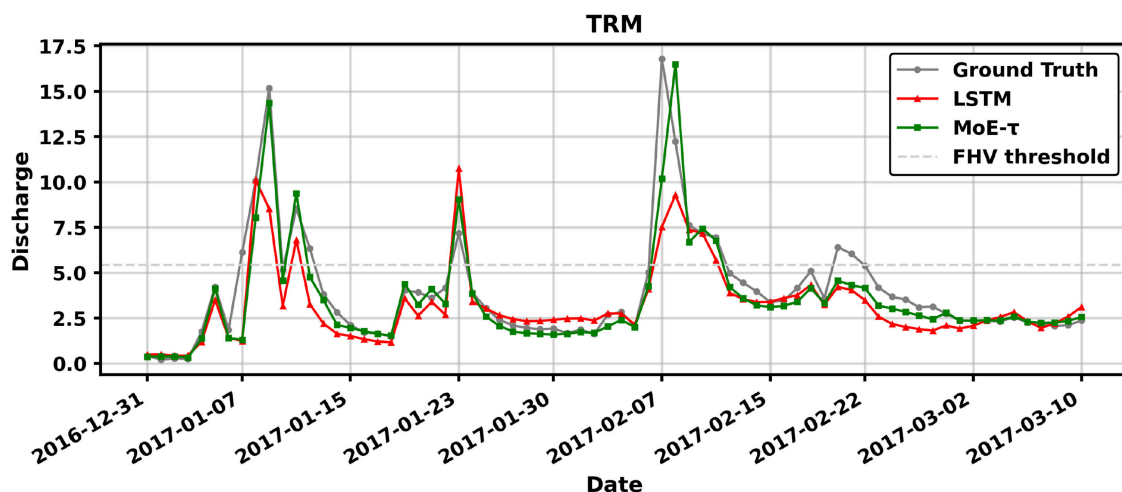


Fig. D.21. Visualization comparison of LSTM and MoE- τ discharge predictions for the TRM watershed over the period 2016/12/31 ~ 2017/03/10.

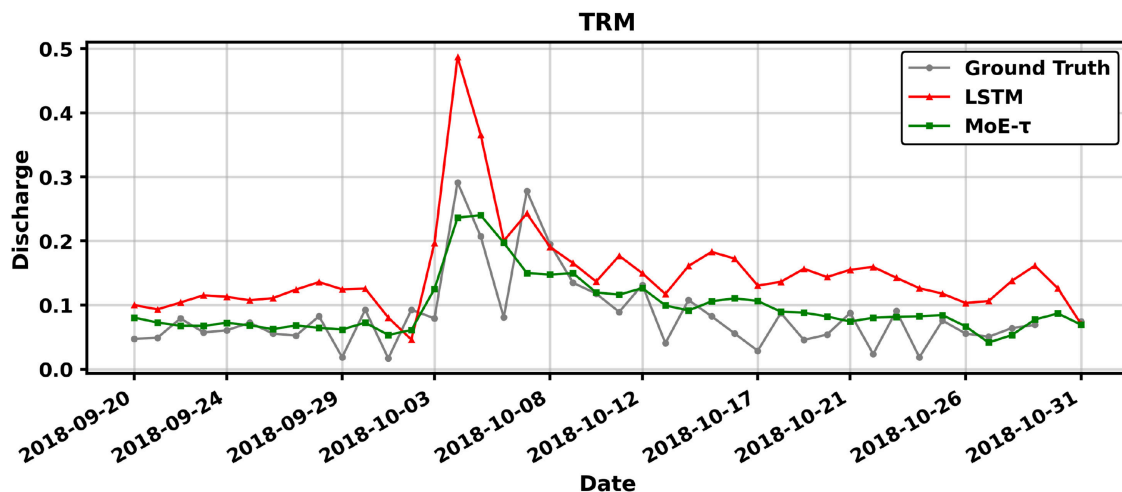


Fig. D.22. Visualization comparison of LSTM and MoE- τ discharge predictions for the TRM watershed over the period 2018/09/20 ~ 2018/10/31.

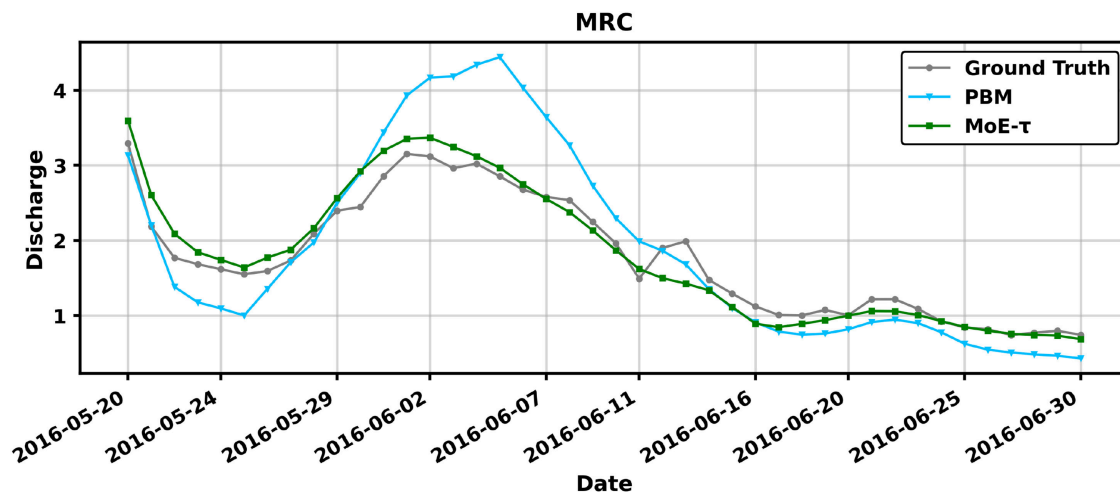


Fig. D.23. Visualization comparison of PBM and MoE- τ discharge predictions for the MRC watershed over the period 2016/05/20 ~ 2016/06/30.

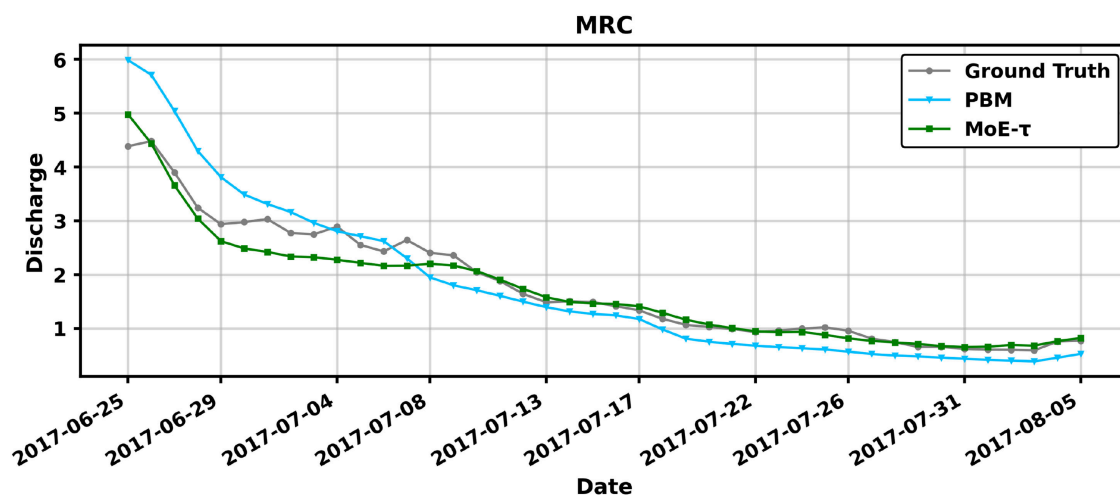


Fig. D.24. Visualization comparison of PBM and MoE- τ discharge predictions for the MRC watershed over the period 2017/06/25 ~ 2017/08/05.

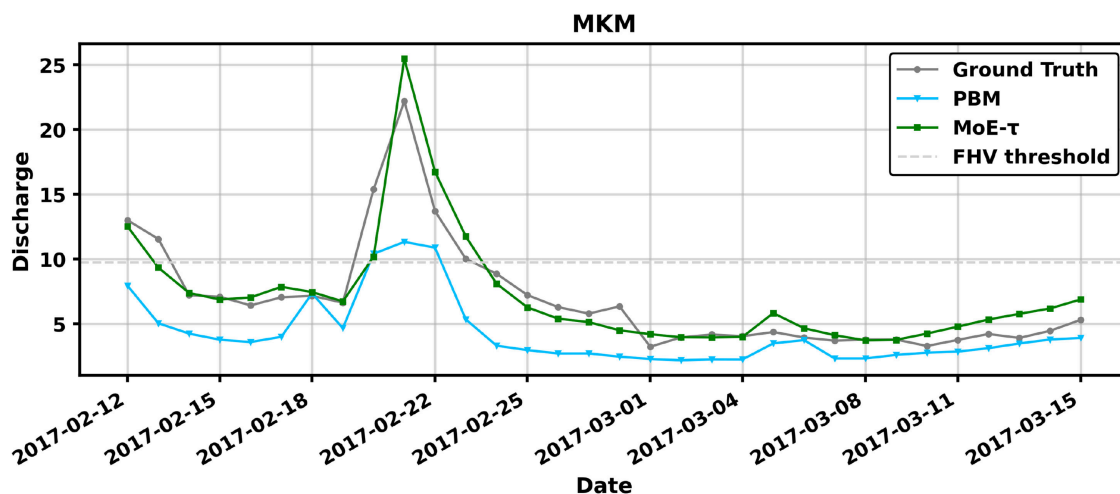


Fig. D.25. Visualization comparison of PBM and MoE- τ discharge predictions for the MKM watershed over the period 2017/02/12 ~ 2017/03/15.

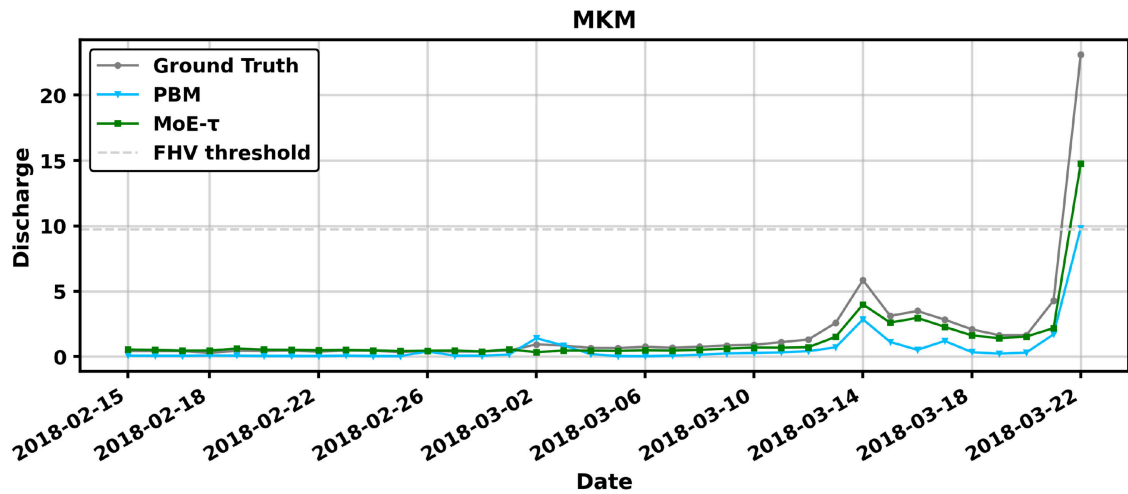


Fig. D.26. Visualization comparison of PBM and MoE- τ discharge predictions for the MKM watershed over the period 2018/02/15 ~ 2018/03/22.

Appendix E. Hydrograph comparison across all watersheds for 2015 and 2017

Figs. E.27–E.52.

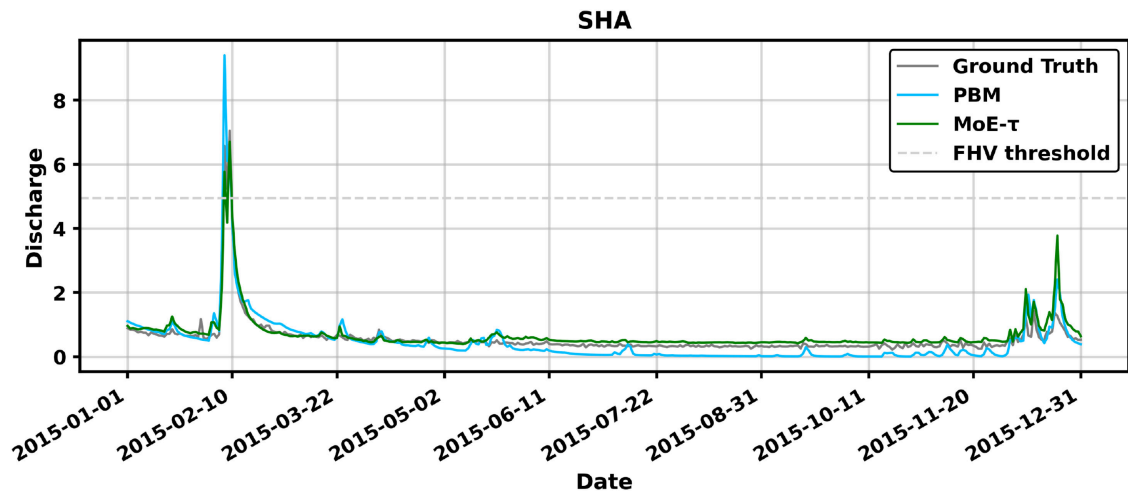


Fig. E.27. Visualization comparison of PBM and MoE- τ discharge predictions for the SHA watershed over the 2015 testing period.

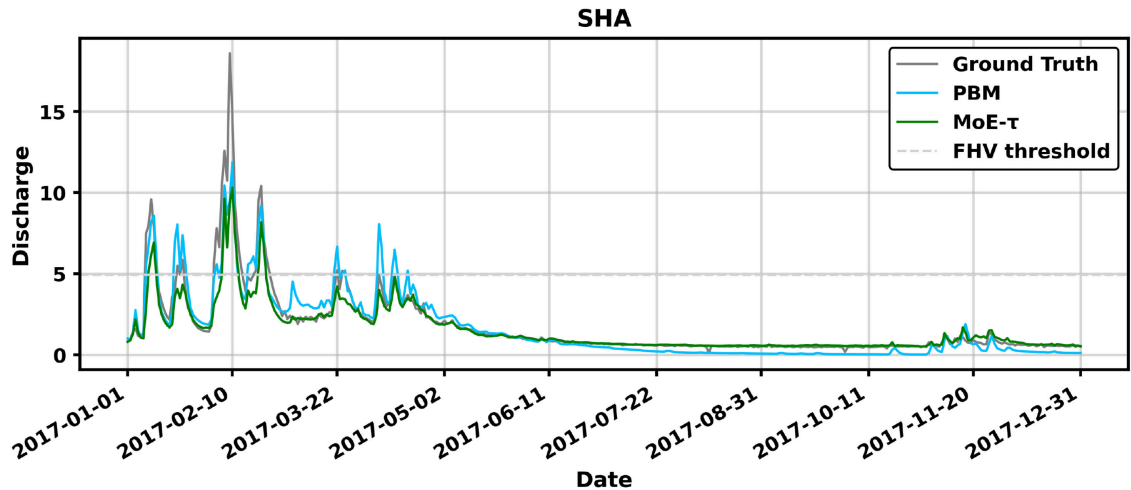


Fig. E.28. Visualization comparison of PBM and MoE- τ discharge predictions for the SHA watershed over the 2017 testing period.

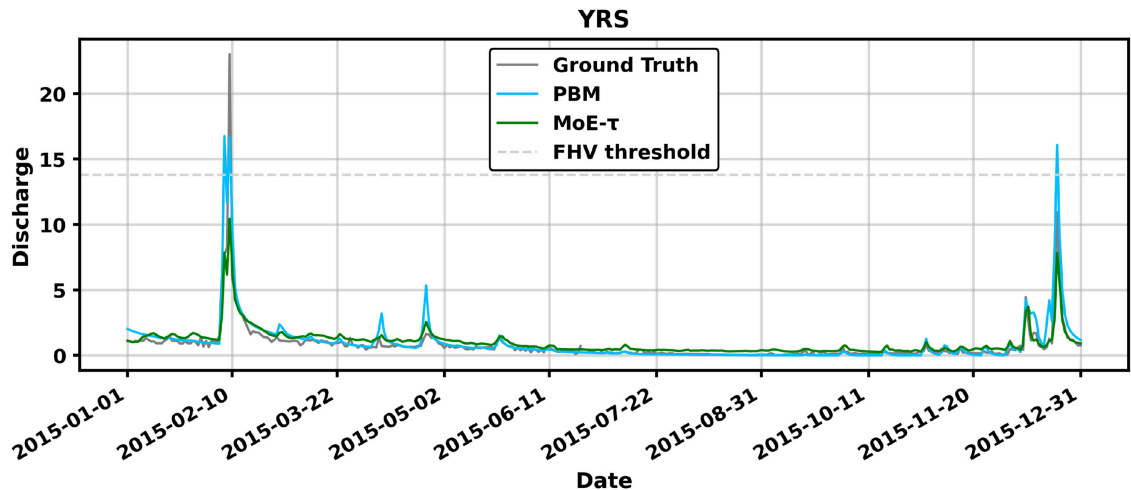


Fig. E.29. Visualization comparison of PBM and MoE- τ discharge predictions for the YRS watershed over the 2015 testing period.

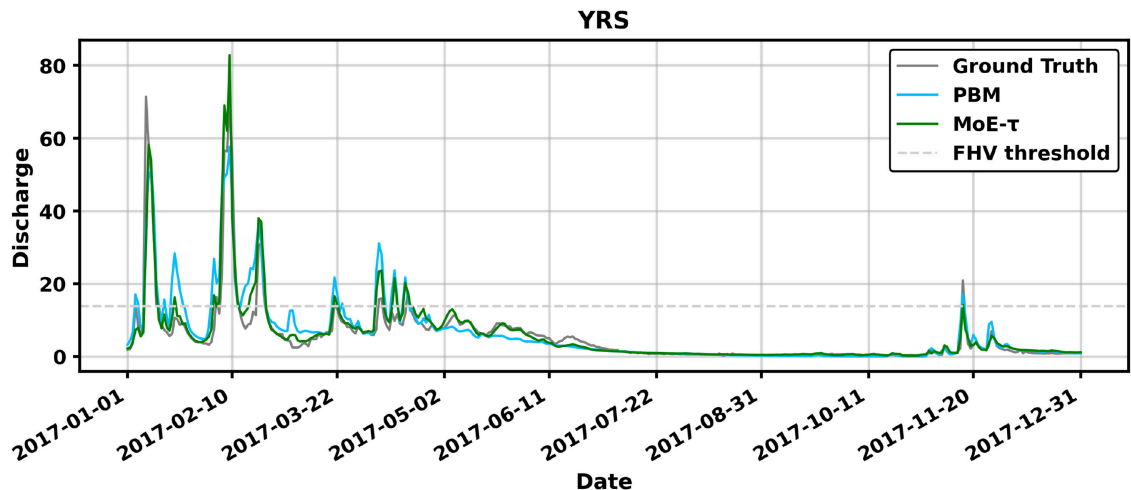


Fig. E.30. Visualization comparison of PBM and MoE- τ discharge predictions for the YRS watershed over the 2017 testing period.

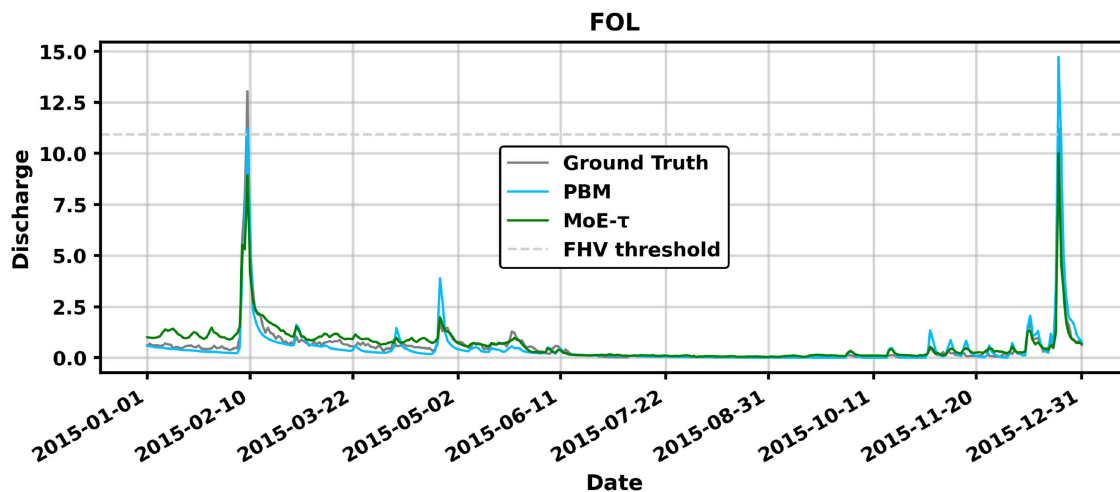


Fig. E.31. Visualization comparison of PBM and MoE- τ discharge predictions for the FOL watershed over the 2015 testing period.

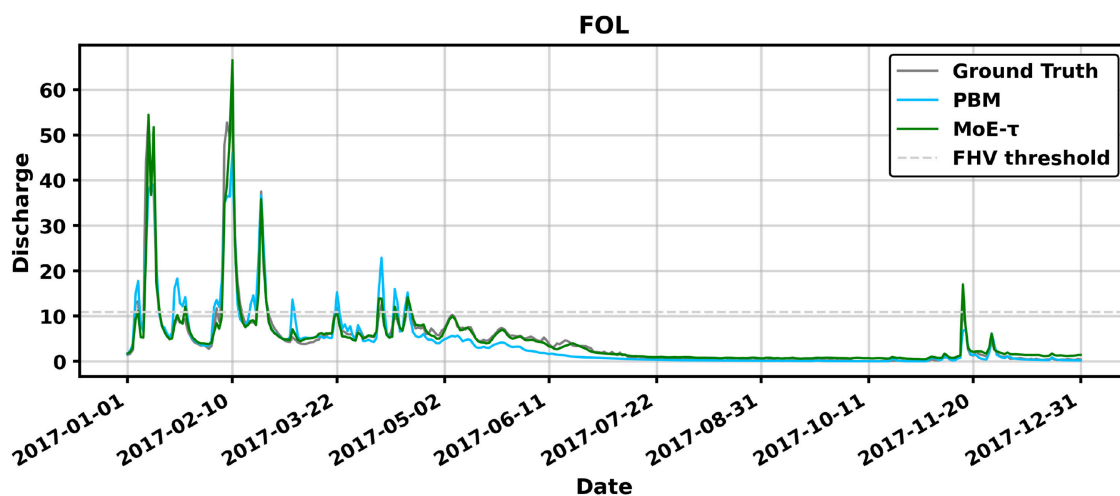


Fig. E.32. Visualization comparison of PBM and MoE- τ discharge predictions for the FOL watershed over the 2017 testing period.

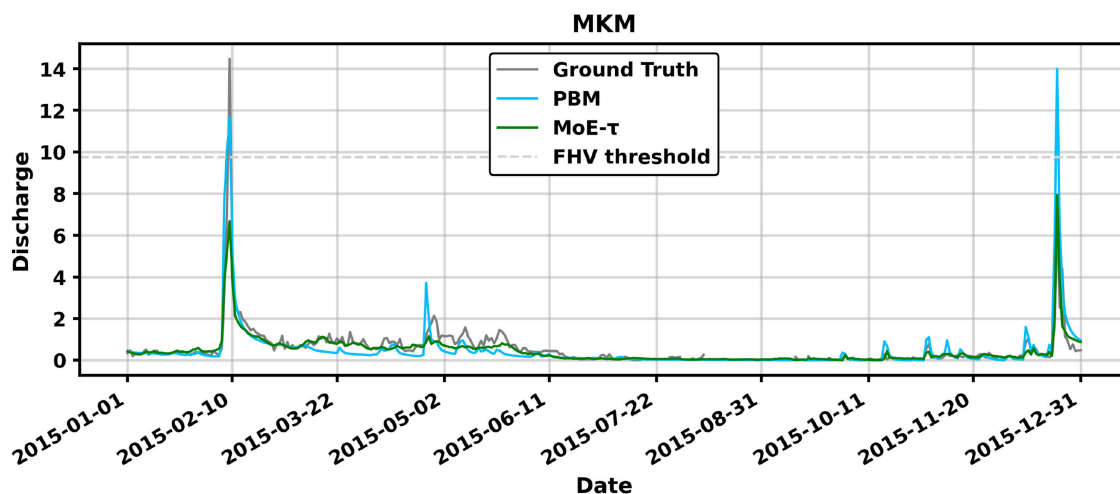


Fig. E.33. Visualization comparison of PBM and MoE- τ discharge predictions for the MKM watershed over the 2015 testing period.

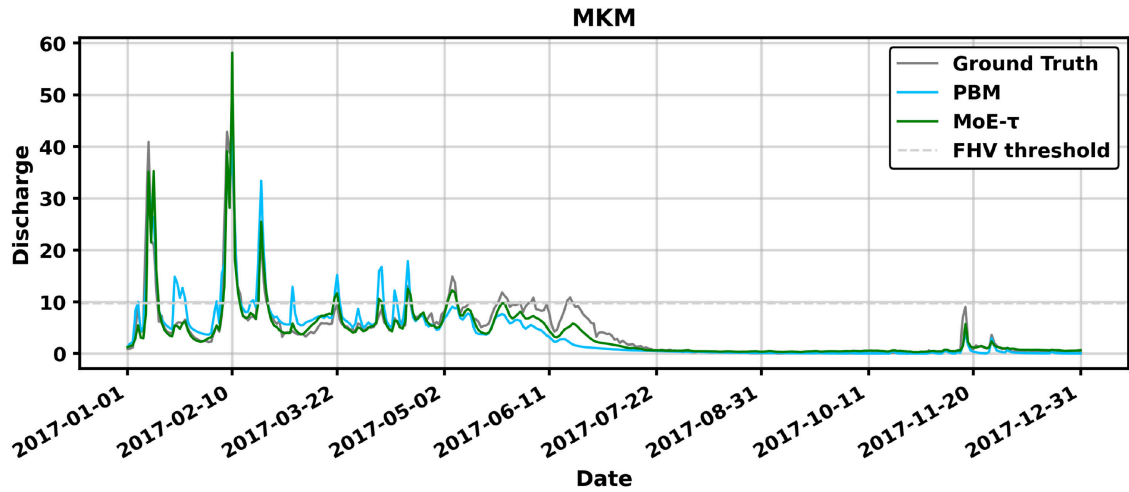


Fig. E.34. Visualization comparison of PBM and MoE- τ discharge predictions for the MKM watershed over the 2017 testing period.

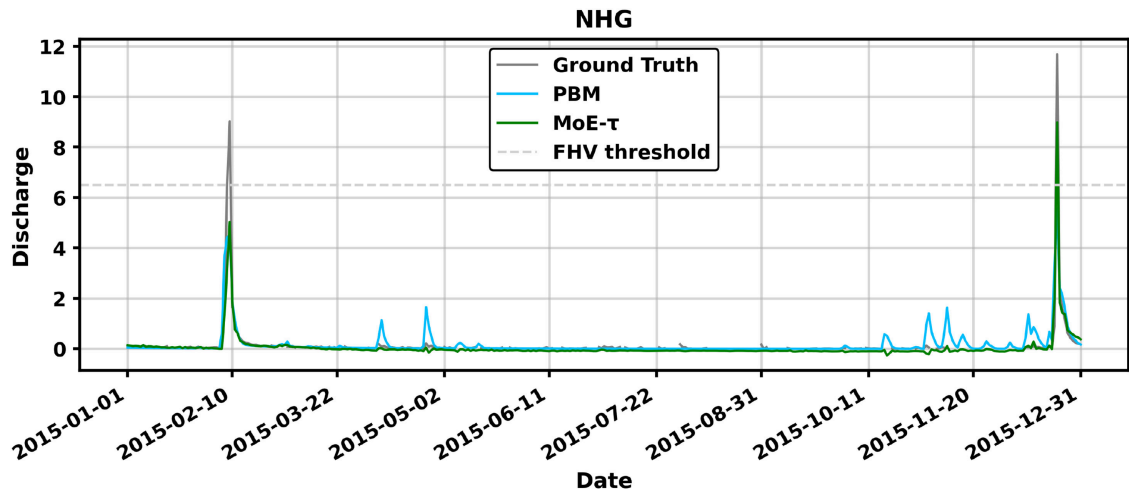


Fig. E.35. Visualization comparison of PBM and MoE- τ discharge predictions for the NHG watershed over the 2015 testing period.

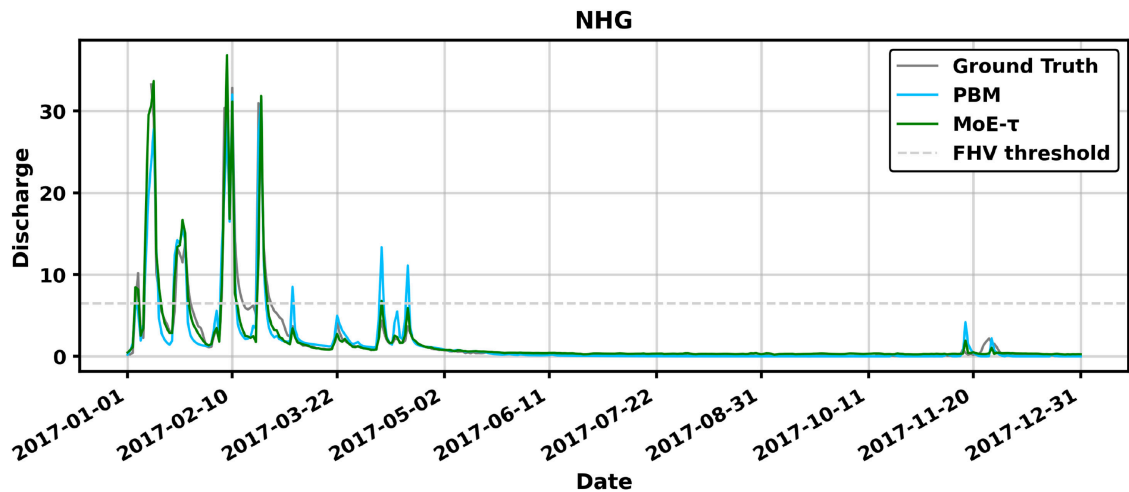


Fig. E.36. Visualization comparison of PBM and MoE- τ discharge predictions for the NHG watershed over the 2017 testing period.

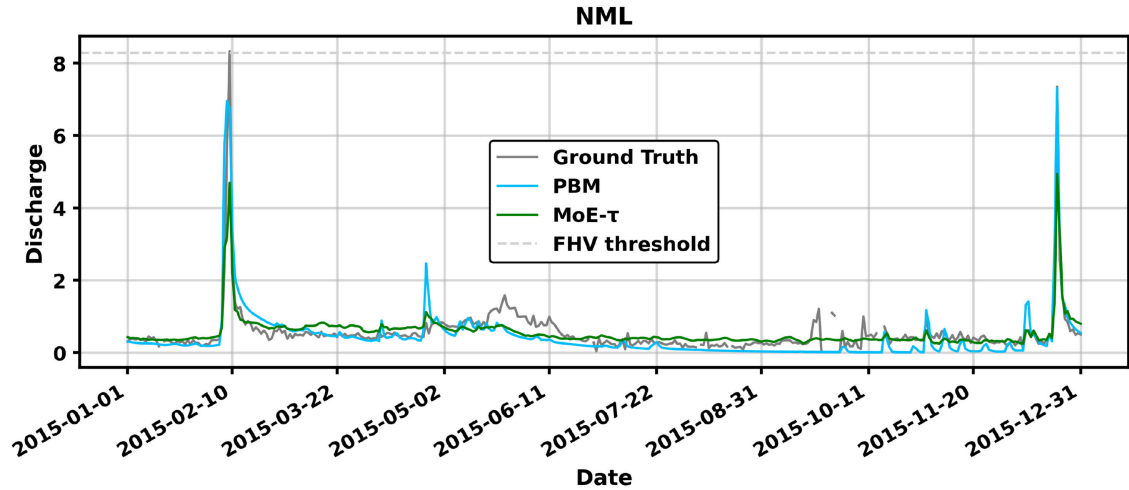


Fig. E.37. Visualization comparison of PBM and MoE- τ discharge predictions for the NML watershed over the 2015 testing period.

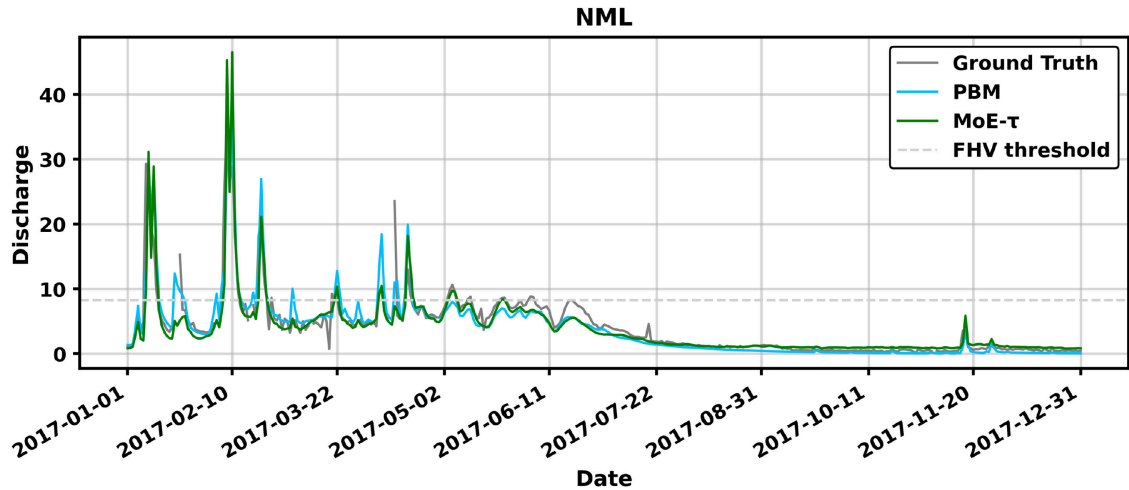


Fig. E.38. Visualization comparison of PBM and MoE- τ discharge predictions for the NML watershed over the 2017 testing period.

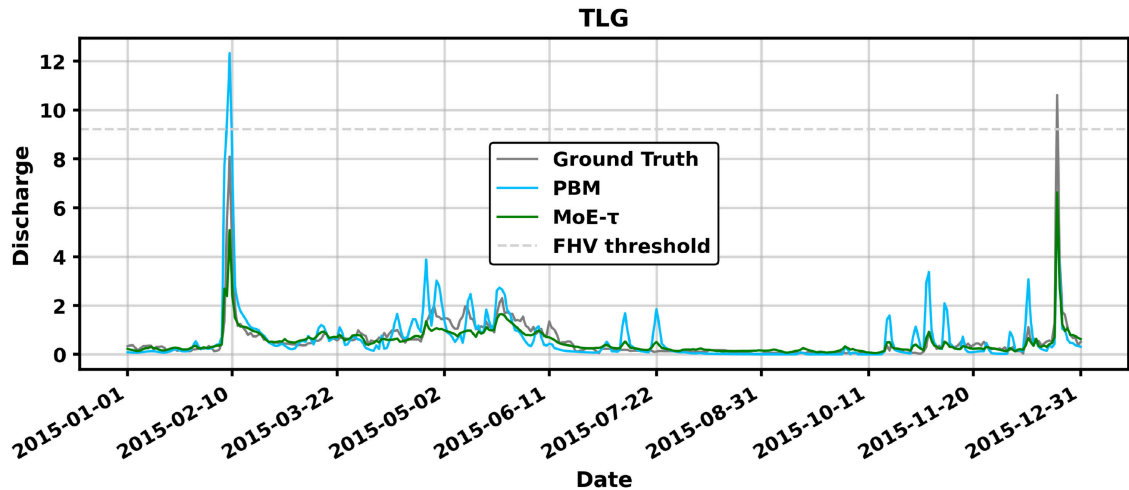


Fig. E.39. Visualization comparison of PBM and MoE- τ discharge predictions for the TLG watershed over the 2015 testing period.

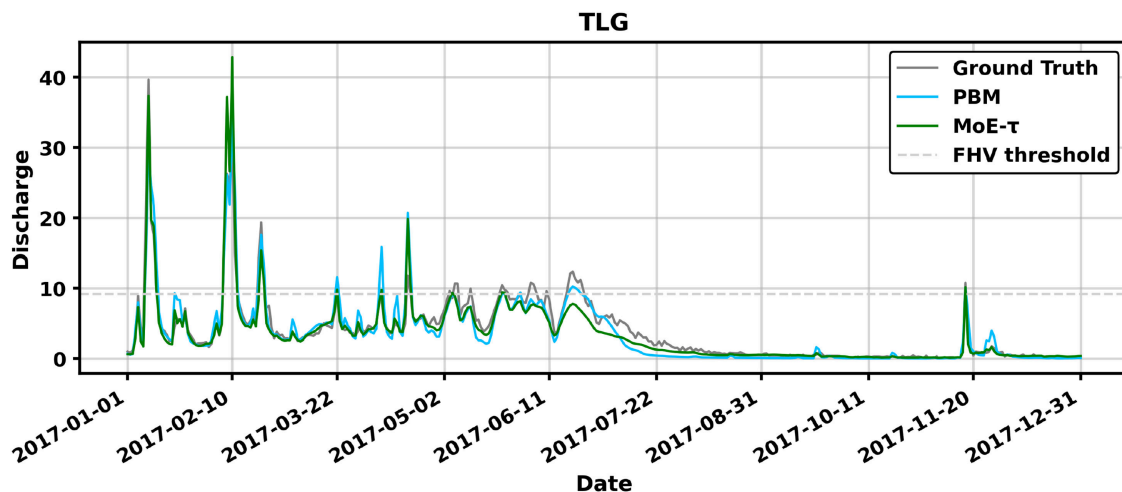


Fig. E.40. Visualization comparison of PBM and MoE- τ discharge predictions for the TLG watershed over the 2017 testing period.

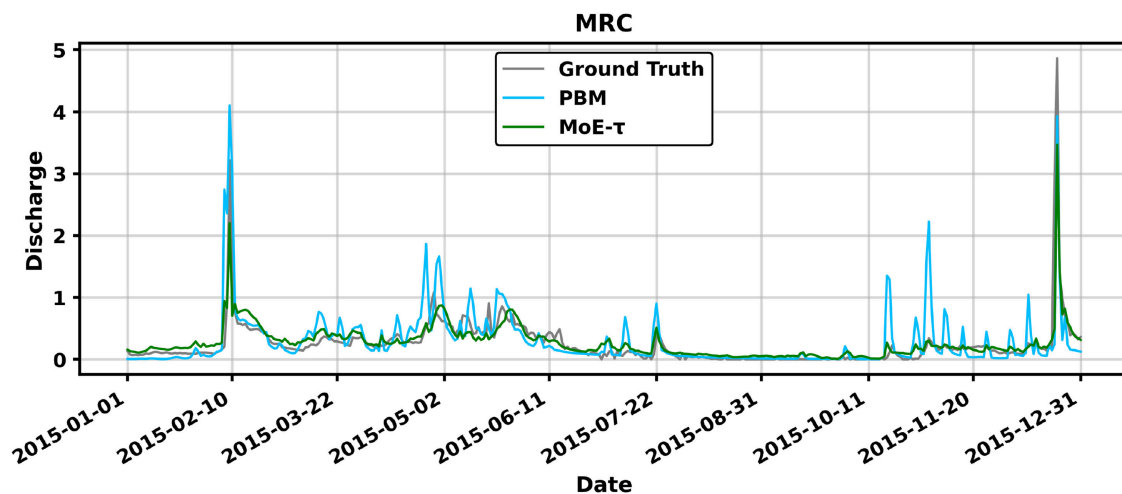


Fig. E.41. Visualization comparison of PBM and MoE- τ discharge predictions for the MRC watershed over the 2015 testing period.

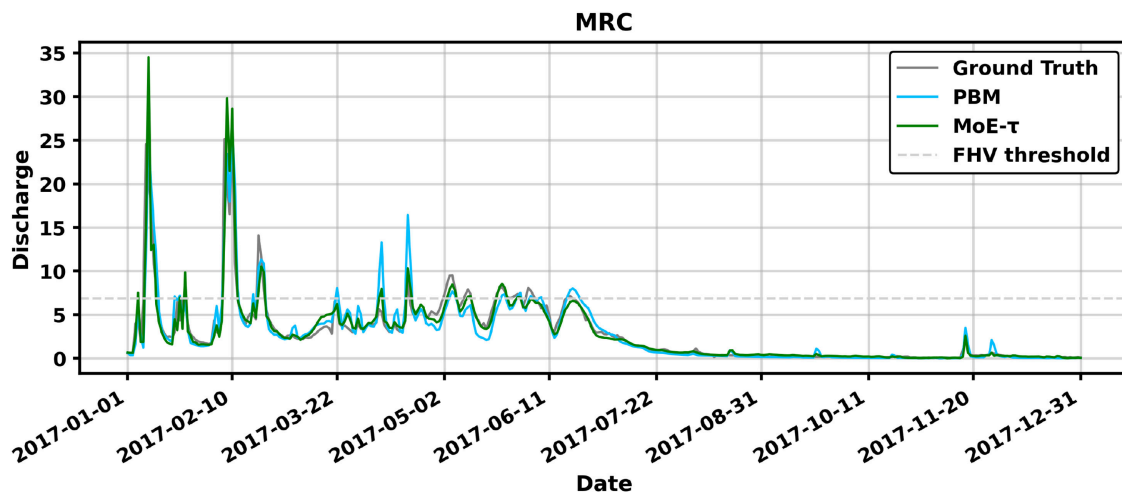


Fig. E.42. Visualization comparison of PBM and MoE- τ discharge predictions for the MRC watershed over the 2017 testing period.

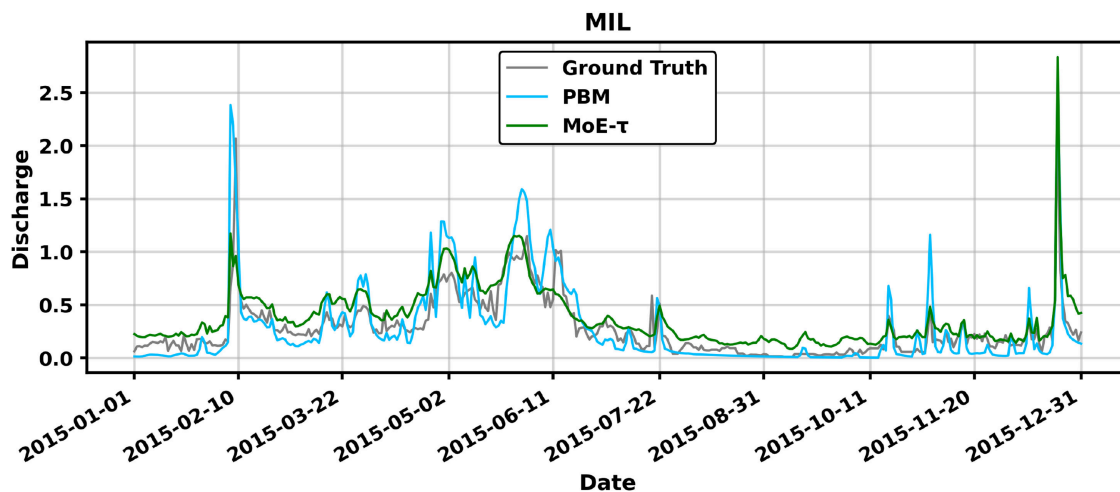


Fig. E.43. Visualization comparison of PBM and MoE- τ discharge predictions for the MIL watershed over the 2015 testing period.

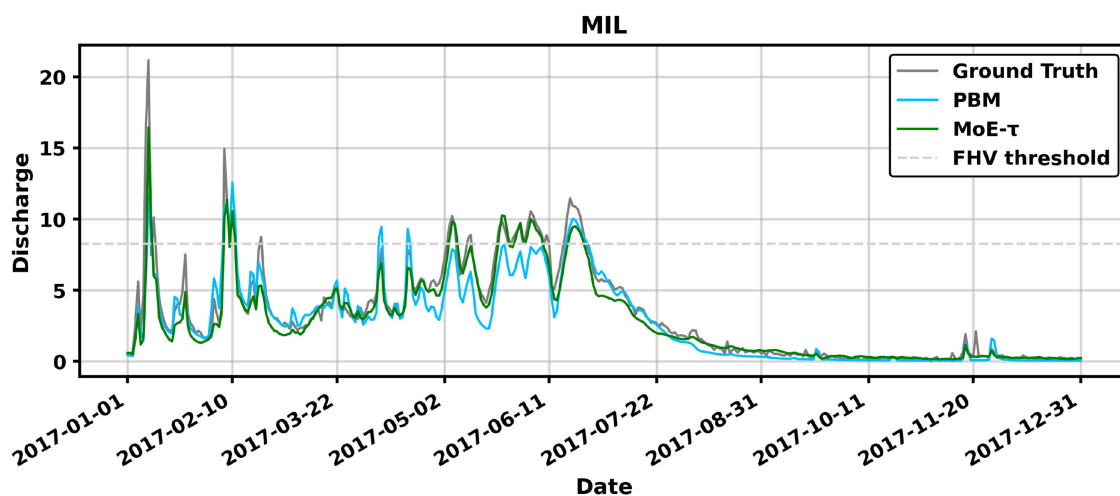


Fig. E.44. Visualization comparison of PBM and MoE- τ discharge predictions for the MIL watershed over the 2017 testing period.

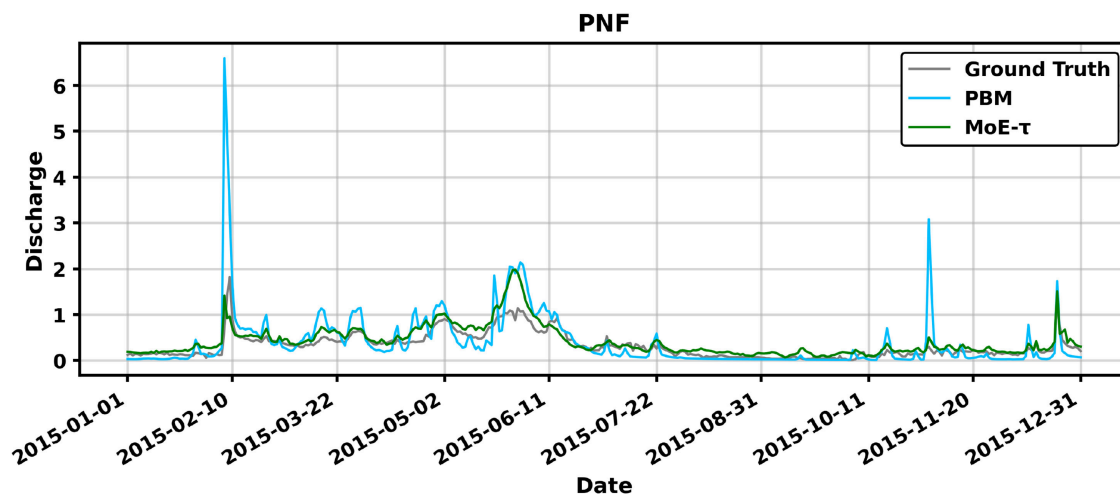


Fig. E.45. Visualization comparison of PBM and MoE- τ discharge predictions for the PNF watershed over the 2015 testing period.

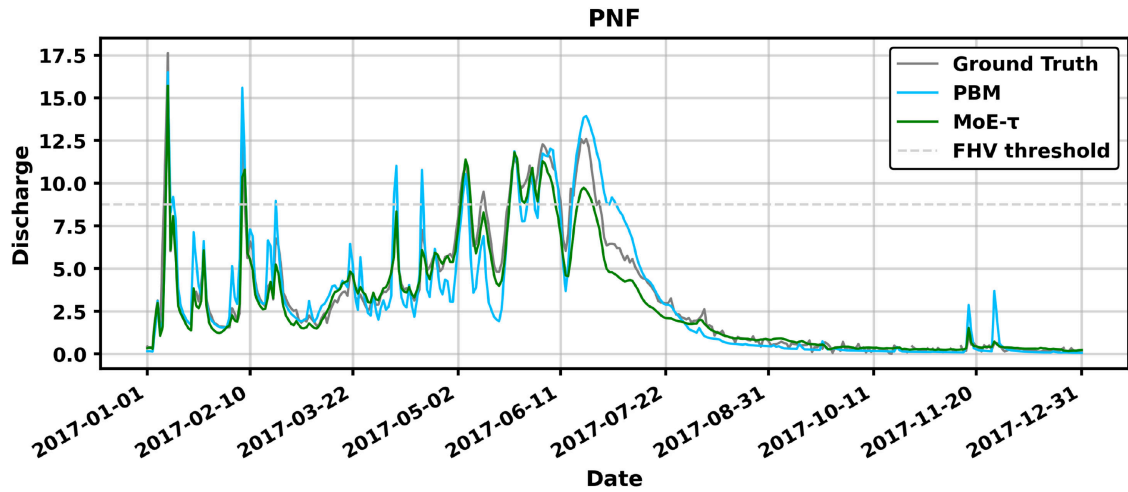


Fig. E.46. Visualization comparison of PBM and MoE- τ discharge predictions for the PNF watershed over the 2017 testing period.

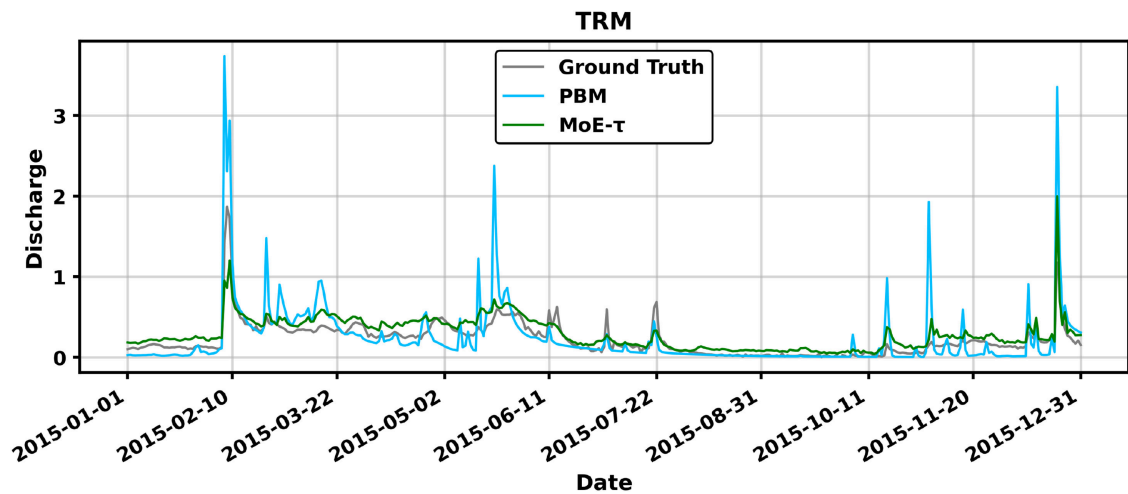


Fig. E.47. Visualization comparison of PBM and MoE- τ discharge predictions for the TRM watershed over the 2015 testing period.

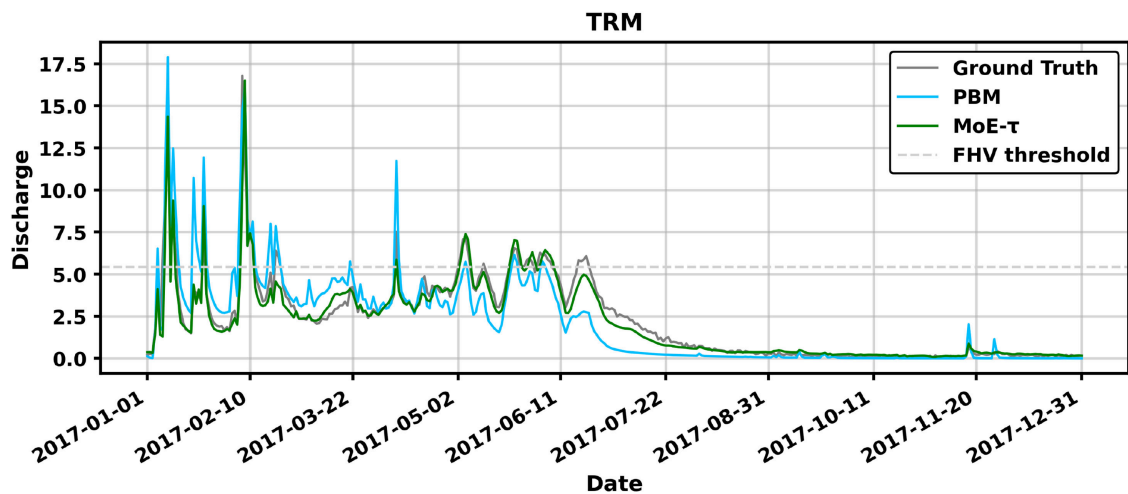


Fig. E.48. Visualization comparison of PBM and MoE- τ discharge predictions for the TRM watershed over the 2017 testing period.

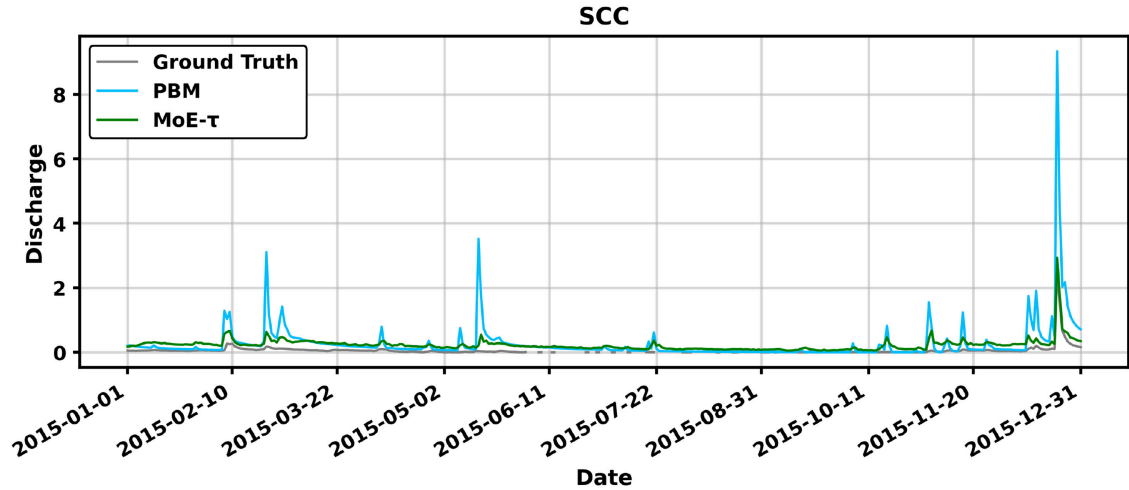


Fig. E.49. Visualization comparison of PBM and MoE- τ discharge predictions for the SCC watershed over the 2015 testing period.

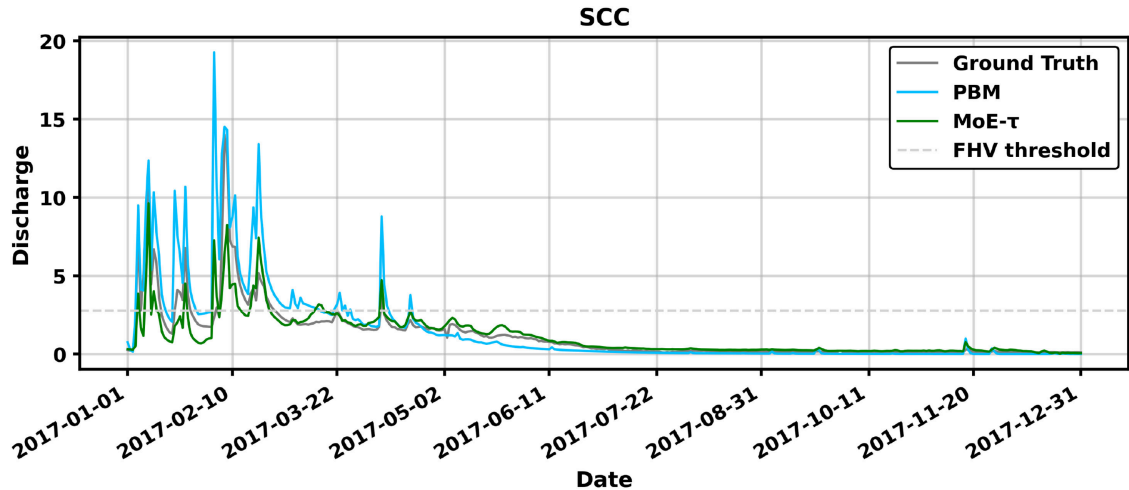


Fig. E.50. Visualization comparison of PBM and MoE- τ discharge predictions for the SCC watershed over the 2017 testing period.

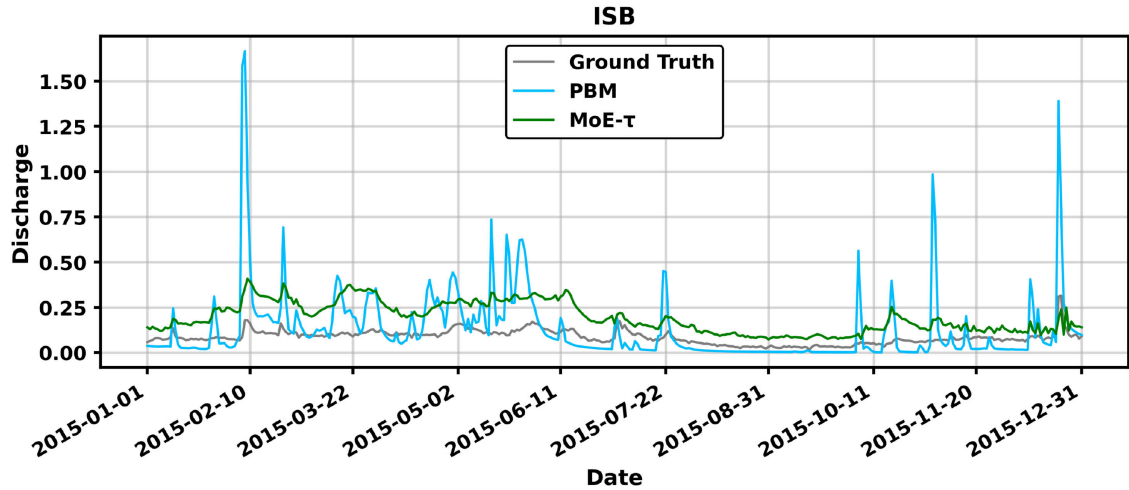


Fig. E.51. Visualization comparison of PBM and MoE- τ discharge predictions for the ISB watershed over the 2015 testing period.

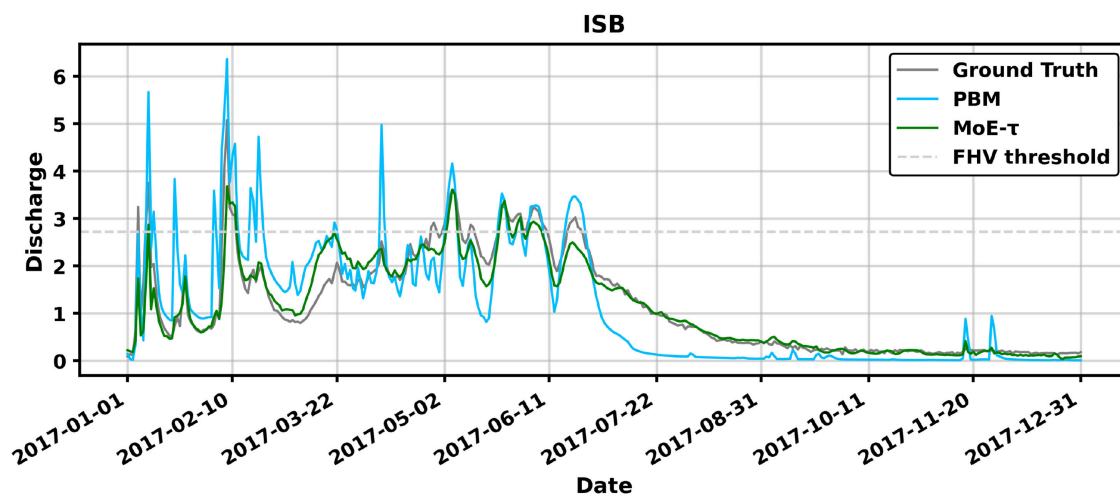


Fig. E.52. Visualization comparison of PBM and MoE- τ discharge predictions for the ISB watershed over the 2017 testing period.

References

- Adera, S., Bellugi, D., Dhakal, A., & Larsen, L. (2024). Streamflow prediction at the intersection of physics and machine learning: A case study of two mediterranean-climate watersheds. *Water Resources Research*, 60(7), e2023WR035790.
- Aguayo, R., Zekollari, H., Hanus, S., Baez-Villanueva, O. M., Mendoza, P. A., & Maussion, F. (2025). Hybrid glacio-hydrological modeling reveals contrasting runoff changes in western patagonia over the 21st century. *Earth's Future*, 13(11), e2025EF006442.
- An, K. (1933). Sulla determinazione empirica di una legge di distribuzione. *Giorn Dell'inst Ital Degli Att*, 4, 89–91.
- Anderson, E. A. (1973). National weather service river forecast system—snow accumulation and ablation model. Tech. Memo NWS HYDRO-17. NOAA Silver Spring, MD.
- Arsenault, R., Martel, J.-L., Brunet, F., Brissette, F., & Mai, J. (2023). Continuous streamflow prediction in ungauged basins: Long short-term memory neural networks clearly outperform traditional hydrological models. *Hydrology and Earth System Sciences*, 27(1), 139–157.
- Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. arXiv:1409.0473
- Barnett, T. P., Adam, J. C., & Lettenmaier, D. P. (2005). Potential impacts of a warming climate on water availability in snow-dominated regions. *Nature*, 438(7066), 303–309.
- Becker, L., Sparks, D., Fults, D. M., & Yeh, W. W. G. (1976). Operations models for central valley project. *Journal of the Water Resources Planning and Management Division*, 102(1), 101–115.
- Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time series analysis: Forecasting and control*. John Wiley & Sons.
- Burnash, R. J. C. (1973). *A generalized streamflow simulation system: conceptual modeling for digital computers*. US Department of Commerce, National Weather Service, and State of California....
- Burnash, R. J. C. (1995). The NWS river forecast system—catchment modeling. In V. P. Singh (Ed.), *Computer models of watershed hydrology* (pp. 311–366).
- Cho, K., Van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. arXiv:1406.1078
- Das, T., Hidalgo, H. G., Pierce, D. W., Barnett, T. P., Dettinger, M. D., Cayan, D. R., Bonfils, C., Bala, G., & Mirin, A. (2009). Structure and detectability of trends in hydrological measures over the western united states. *Journal of Hydrometeorology*, 10(4), 871–892.
- Delaney, C. J., Hartman, R. K., Mendoza, J., Dettinger, M., Delle Monache, L., Jasperse, J., Ralph, F. M., Talbot, C., Brown, J., Reynolds, D. et al. (2020). Forecast informed reservoir operations using ensemble streamflow predictions for a multipurpose reservoir in Northern California. *Water Resources Research*, 56(9), e2019WR026604.
- Dettinger, M. D., Ralph, F. M., Das, T., Neiman, P. J., & Cayan, D. R. (2011). Atmospheric rivers, floods and the water resources of california. *Water*, 3(2), 445–478.
- Ding, C., Wang, Z., Tan, Z., & Zhang, X. (2026). Multi-source heterogeneous data-driven interpretable model based on transformer and Kolmogorov-Arnold networks for streamflow prediction. *Expert Systems with Applications*, 328, 132862.
- Draper, A. J., Munévar, A., Arora, S. K., Reyes, E., Parker, N. L., Chung, F. I., & Peterson, L. E. (2004). Calsim: Generalized model for reservoir system analysis. *Journal of Water Resources Planning and Management*, 130(6), 480–489.
- Duan, Q., Sorooshian, S., & Gupta, V. (1992). Effective and efficient global optimization for conceptual rainfall-runoff models. *Water Resources Research*, 28(4), 1015–1031.
- Fang, Y., Liu, Y., & Margulis, S. A. (2022). Western United States UCLA daily snow reanalysis, version 1. [Data set]. Accessed via NSIDC DAAC. <https://doi.org/10.5067/PP7T2GBI52I2>
- Frame, J., Kratzert, F., Klotz, D., Gauch, M., Shelev, G., Gilon, O., Qualls, L. M., Gupta, H. V., & Nearing, G. S. (2021a). Deep learning rainfall-runoff predictions of extreme events. *Hydrology and Earth System Sciences Discussions*, 2021, 1–20.
- Frame, J. M., Kratzert, F., Raney, A., Rahman, M., Salas, F. R., & Nearing, G. S. (2021b). Post-processing the national water model with long short-term memory networks for streamflow predictions and model diagnostics. *JAWRA Journal of the American Water Resources Association*, 57(6), 885–905.
- Godsey, S. E., Kirchner, J. W., & Tague, C. L. (2014). Effects of changes in winter snowpacks on summer low flows: Case studies in the Sierra Nevada, California, USA. *Hydrological Processes*, 28(19), 5048–5064.
- Graves, A. (2012). Long short-term memory. *Supervised Sequence Labelling With Recurrent Neural Networks*, 385, 37–45.
- Gupta, H. V., Kling, H., Yilmaz, K. K., & Martinez, G. F. (2009). Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *Journal of Hydrology*, 377(1–2), 80–91.
- Hamon, W. R. (1961). Estimating potential evapotranspiration. *Journal of the Hydraulics Division, ASCE*, 87, 107–120.
- Hansen, M. C., Defries, R. S., Townshend, J. R. G., & Sohlberg, R. (2010). Global land cover classification at 1 km spatial resolution using a classification tree approach. *International Journal of Remote Sensing*, 21(6–7), 1331–1364. <https://doi.org/10.1080/014311600210209>
- He, M., Russo, M., & Anderson, M. (2017). Hydroclimatic characteristics of the 2012–2015 california drought from an operational perspective. *Climate*, 5(1), 5.

- He, M., Whitin, B., Hartman, R., Henkel, A., Fickenschers, P., Staggs, S., Morin, A., Imgarten, M., Haynes, A., & Russo, M. (2016). Verification of ensemble water supply forecasts for sierra nevada watersheds. *Hydrology*, 3(4), 35.
- He, Y., Wang, Z., Cheng, H., & Ding, W. (2026). Multi-scale feature fusion and uncertainty quantification in streamflow prediction: A temporal convolutional network approach with hybrid denoising. *Environmental Modelling & Software*, 198, 106879.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.
- Hogue, T. S., Sorooshian, S., Gupta, H., Holz, A., & Braatz, D. (2000). A multistep automatic calibration scheme for river forecasting models. *Journal of Hydrometeorology*, 1(6), 524–542.
- Hsu, K.-I., Gupta, H. V., & Sorooshian, S. (1995). Artificial neural network modeling of the rainfall-runoff process. *Water Resources Research*, 31(10), 2517–2530.
- Hunt, K. M. R., Matthews, G. R., Pappenberger, F., & Prudhomme, C. (2022). Using a long short-term memory (LSTM) neural network to boost river streamflow forecasts over the western united states. *Hydrology and Earth System Sciences*, 26(21), 5449–5472.
- Jacobs, R. A., Jordan, M. I., Nowlan, S. J., & Hinton, G. E. (1991). Adaptive mixtures of local experts. *Neural Computation*, 3(1), 79–87.
- Jarvis, A., Reuter, H. I., Nelson, A., & Guevara, E. (2008). Hole-filled SRTM for the globe version 4. Available from the CGIAR-CSI SRTM 90m Database (<http://srtm.csi.cgiar.org>).
- Kapnick, S., & Hall, A. (2010). Observed climate–snowpack relationships in California and their implications for the future. *Journal of Climate*, 23(13), 3446–3456.
- Kim, S.-H., Hwang, J., Devineni, N., & Sankarasubramanian, A. (2026). Do deep-learning models perform better than process-based models? A diagnostic evaluation using synthetic streamflow simulation under various sources of uncertainties. *Water Resources Research*, 62(5), e2025WR040439.
- Kratzert, F., Gauch, M., Nearing, G., & Klotz, D. (2022). Neuralhydrology—a python library for deep learning research in hydrology. *Journal of Open Source Software*, 7(71), 4050. <https://doi.org/10.21105/joss.04050>
- Kratzert, F., Klotz, D., Brenner, C., Schulz, K., & Herrnegger, M. (2018). Rainfall–runoff modelling using long short-term memory (LSTM) networks. *Hydrology and Earth System Sciences*, 22(11), 6005–6022.
- Kratzert, F., Klotz, D., Hochreiter, S., & Nearing, G. S. (2021). A note on leveraging synergy in multiple meteorological data sets with deep learning for rainfall–runoff modeling. *Hydrology and Earth System Sciences*, 25, 2685–2703. <https://doi.org/10.5194/hess-25-2685-2021>
- Kratzert, F., Klotz, D., Shalev, G., Klambauer, G., Hochreiter, S., & Nearing, G. (2019). Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets. *Hydrology and Earth System Sciences*, 23(12), 5089–5110.
- Liu, J., Shen, C., O'Donncha, F., Song, Y., Zhi, W., Beck, H. E., Bindas, T., Kraabel, N., & Lawson, K. (2025). From RNNs to transformers: Benchmarking deep learning architectures for hydrologic prediction. *Hydrology and Earth System Sciences*, 29(23), 6811–6828.
- Lohmann, D., Raschke, E., Nijssen, B., & Lettenmaier, D. (1998). Regional scale hydrology: I. Formulation of the VIC-2L model coupled to a routing model. *Hydrological Sciences Journal*, 43(1), 131–141. <https://doi.org/10.1080/02626669809492107>
- Luce, C. H., & Holden, Z. A. (2009). Declining annual streamflow distributions in the Pacific Northwest United States, 1948–2006. *Geophysical Research Letters*, 36(16).
- Lynn, E., Cuthbertson, A., He, M., Vasquez, J. P., Anderson, M. L., Coombe, P., Abatzoglou, J. T., & Hatchett, B. J. (2020). Precipitation-phase partitioning at landscape scales to regional scales. *Hydrology and Earth System Sciences*, 24(11), 5317–5328.
- Miller, D., & White, R. A. (1998). A continuous United States multilayer soil characteristics dataset for regional climate and hydrology modeling. *Earth Interactions*, 2(2), 1–26. [https://doi.org/10.1175/1087-3562\(1998\)002<0001:ACUSMS>2.3.CO;2](https://doi.org/10.1175/1087-3562(1998)002<0001:ACUSMS>2.3.CO;2)
- Milly, P. C. D., Dunne, K. A., & Vecchia, A. V. (2005). Global pattern of trends in streamflow and water availability in a changing climate. *Nature*, 438(7066), 347–350.
- Molina, A. R., Frame, J. M., Hलगren, J., & Gong, J. (2025). A proof of concept for improving estimates of ungauged basin streamflow via an LSTM-based synthetic network simulation approach. *Journal of Geophysical Research: Machine Learning and Computation*, 2, e2024JH000405.
- Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models Part I - A discussion of principles. *Journal of Hydrology*, 10(3), 282–290. [https://doi.org/10.1016/0022-1694\(70\)90255-6](https://doi.org/10.1016/0022-1694(70)90255-6)
- Oki, T., & Kanae, S. (2006). Global hydrological cycles and world water resources. *Science*, 313(5790), 1068–1072.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, J., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J., & Chintala, S. (2019). Pytorch: An imperative style, high-performance deep learning library. In *Advances in neural information processing systems*. (vol. 32).
- Rhoades, A. M., Jones, A. D., & Ullrich, P. A. (2018). The changing character of the California Sierra Nevada as a natural reservoir. *Geophysical Research Letters*, 45(23), 13–008.
- Rong, Z., Sun, W., Xie, Y., Huang, Z., & Chen, X. (2025). Mixture of experts leveraging informer and LSTM variants for enhanced daily streamflow forecasting. *Journal of Hydrology*, 653, 132737.
- Sabet, M. H., & Coe, J. Q. (1986). Models for water and power scheduling for the california state water project 1. *JAWRA Journal of the American Water Resources Association*, 22(4), 587–596.
- Schuster, M., & Paliwal, K. K. (1997). Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, 45(11), 2673–2681.
- Shi, X., Chen, Z., Wang, H., Yeung, D.-Y., Wong, W.-K., & Woo, W.-c. (2015). Convolutional LSTM network: A machine learning approach for precipitation nowcasting. *Advances in Neural Information Processing Systems*, 28.
- Smirnov, N. V. (1939). Estimate of deviation between empirical distribution functions in two independent samples. *Bulletin Moscow University*, 2(2), 3–16.
- Spearman, C. (1961). The proof and measurement of association between two things.
- Viessman, W., Lewis, G. L., Knapp, J. W., & Harbaugh, T. E. (1989). Introduction to hydrology.
- Vörösmarty, C. J., McIntyre, P. B., Gessner, M. O., Dudgeon, D., Prusevich, A., Green, P., Glidden, S., Bunn, S. E., Sullivan, C. A., Liermann, C. R. et al. (2010). Global threats to human water security and river biodiversity. *Nature*, 467(7315), 555–561.
- Wan, Z. (2014). New refinements and validation of the collection-6 MODIS land-surface temperature/emissivity product. *Remote Sensing of Environment*, 140, 36–45. <https://doi.org/10.1016/j.rse.2013.08.027>
- Wang, Q. J. (1991). The genetic algorithm and its application to calibrating conceptual rainfall-runoff models. *Water Resources Research*, 27(9), 2467–2471. <https://doi.org/10.1029/91WR01305>
- Wi, S., & Steinschneider, S. (2022). Assessing the physical realism of deep learning hydrologic model projections under climate change. *Water Resources Research*, 58, e2022WR032123. <https://doi.org/10.1029/2022WR032123>
- Wi, S., Yang, Y. C. E., Steinschneider, S., Khalil, A., & Brown, C. M. (2015). Calibration approaches for distributed hydrologic models in poorly gaged basins: Implication for streamflow predictions under climate change. *Hydrology and Earth System Sciences*, 19, 857–876. <https://doi.org/10.5194/hess-19-857-2015>
- Wilbrand, K., Taormina, R., ten Veldhuis, M.-C., Visser, M., Hrachowitz, M., Nuttall, J., & Dahm, R. (2023). Predicting streamflow with LSTM networks using global datasets. *Frontiers in Water*, 5, 1166124.
- Wu, J., Wang, Z., Dong, J., Cui, X., Tao, S., & Chen, X. (2023). Robust runoff prediction with explainable artificial intelligence and meteorological variables from deep learning ensemble model. *Water Resources Research*, 59(9), e2023WR035676.
- Xiao, M., Pan, M., Yang, Y., Cao, Q., Dixon, T., Lewis, G., Su, L., Hartman, R., DeFlorio, M. J., Kalansky, J. F. et al. (2025). Lstm-based post-processing improves streamflow prediction in the Sierra Nevada. *Authoria Preprints*. <https://doi.org/10.22541/essoar.174534330.09680378/v1>.
- Yang, P., & Chen, D. (2026). Enhancing multi-step ahead daily runoff prediction via hydromoe model with local-global hybrid attention. *Water Resources Management*, 40(4), 156.
- Yang, Y., Pan, M., Feng, D., Xiao, M., Dixon, T., Hartman, R., Shen, C., Song, Y., Sengupta, A., Delle Monache, L. et al. (2025). Improving streamflow simulation through machine learning-powered data integration and its potential for forecasting in the Western US. *Hydrology and Earth System Sciences*, 29(20), 5453–5476.
- Yilmaz, K. K., Gupta, H. V., & Wagener, T. (2008). A process-based diagnostic approach to model evaluation: Application to the NWS distributed hydrologic model. *Water Resources Research*, 44(W09417). <https://doi.org/10.1029/2007WR006716>
- Young, P., & Wallis, S. (1985). Recursive estimation: A unified approach to the identification estimation, and forecasting of hydrological systems. *Applied Mathematics and Computation*, 17(4), 299–334.